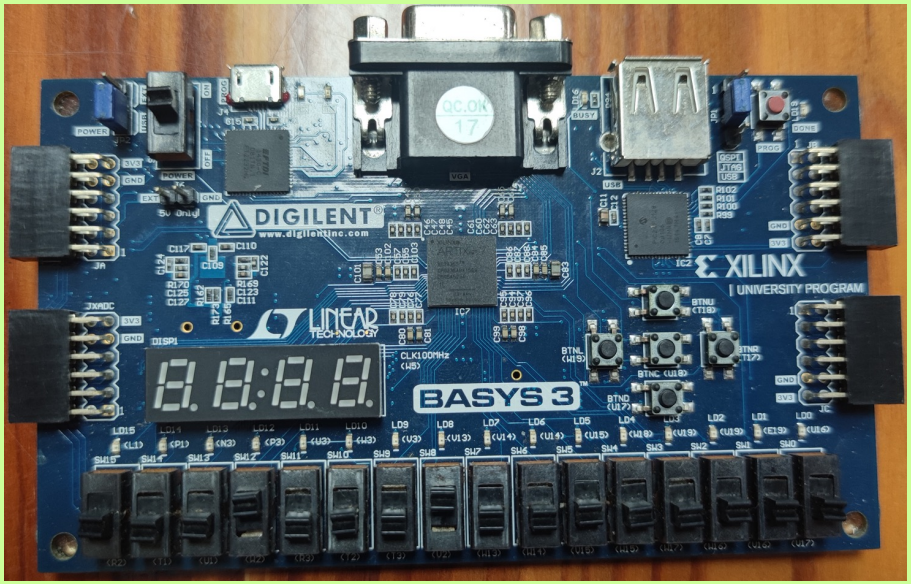
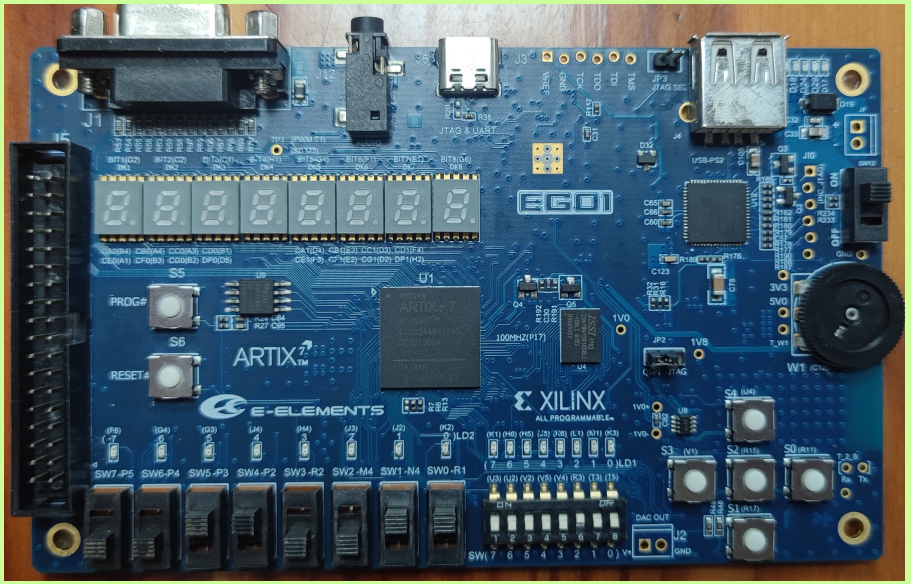
Introduction to Digital System Design

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Play what you like

- Given: Basys3 and EGO1





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What is digital system

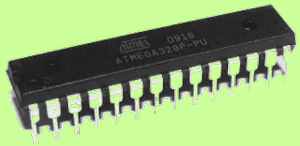
- Everywhere in our life.
- Can be seen as

  “Processor” + “Interface to external”
- Key component: Processor

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What is digital system

- There are two types of microprocessors:
  1. Dedicated microprocessors or microcontrollers
     - Can do only one thing
     - E.g. Traffic light controller



  2. General-purposed microprocessors
     - Can do many things by executing different programs
     - E.g. Intel Core i7 CPU can be programmed to do different things



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From on-off to meaningful task

- A microprocessor is an electronic digital logic circuit implemented inside an integrated circuit (IC) chip
- At the lowest level, it can only understand whether there is electricity or no electricity
- We use 1's and 0's to represent that
- The question is "How do we translate the 1's and 0's into a microprocessor circuit that can do something meaningful?"

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Analog vs Digital

www.mda.gov.sg/PublicEducation/DigitalTvConsumers/Pages/Overview.aspx



Enjoy enhanced viewing experience with Digital TV

website.

Worldwide Digital TV Trends

Why Switch to Digital Broadcasting

Content is increasingly being produced in digital format. Singapore needs to switch to full digital broadcasting in the years to come so as to ensure that consumers can continue to enjoy their favourite TV programmes from around the world. Our industry also needs to keep up-to-date with worldwide digital broadcasting developments to capitalise on the opportunities in digital content and services.

Benefits to Consumers

Digital TV offers:

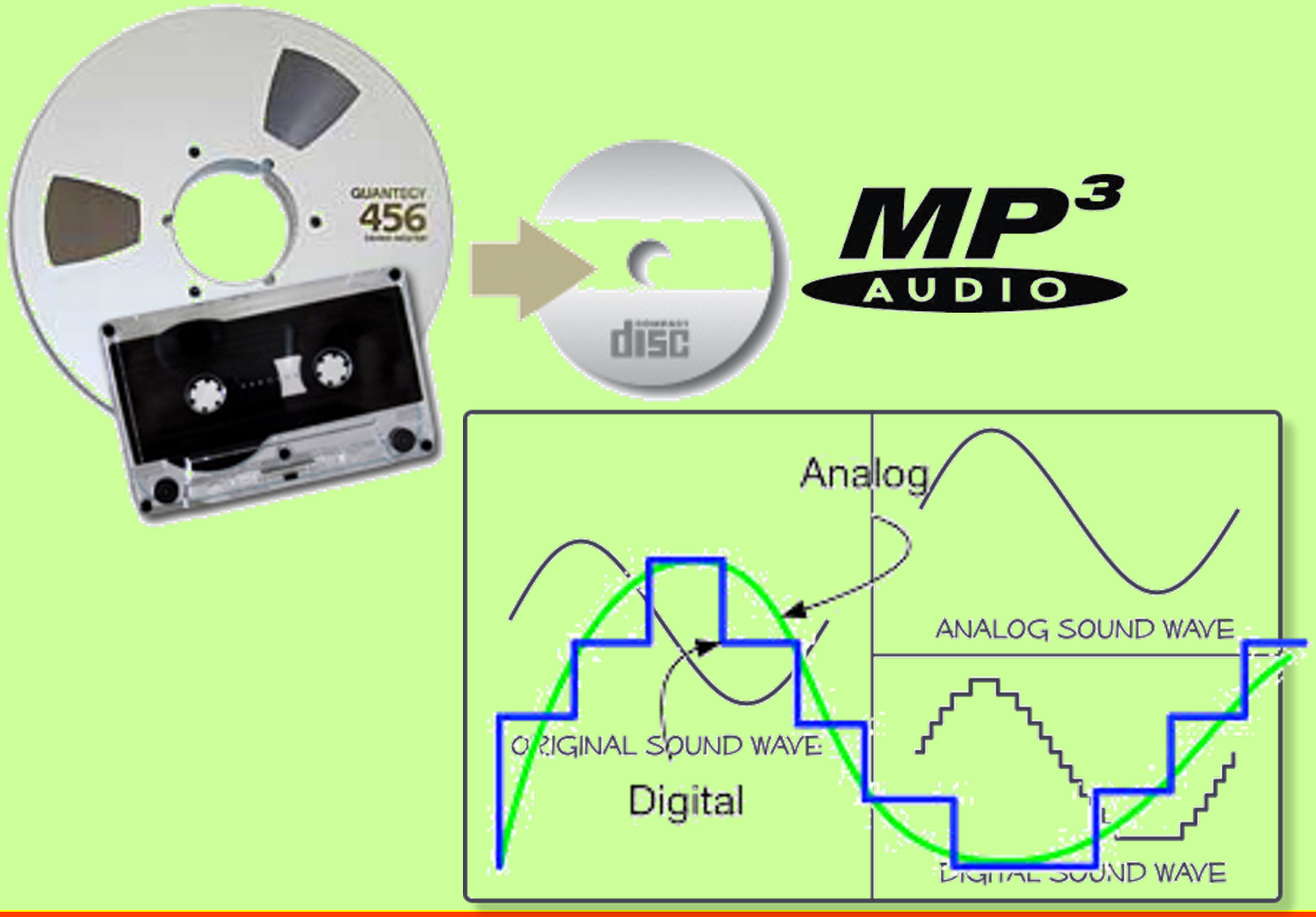
- Better quality pictures (e.g. high definition TV);
- Superior sound (e.g. surround sound);
- Multi-language subtitles; and
- Electronic programme guides which will contain more information about TV programmes

Benefits to Industry

Digital broadcasting uses fewer frequencies than analogue broadcasting. Thus, as Singapore switches over to digital broadcasting, frequencies that are freed up can be made available for other broadcasting and telecommunications services such as wireless broadband services.

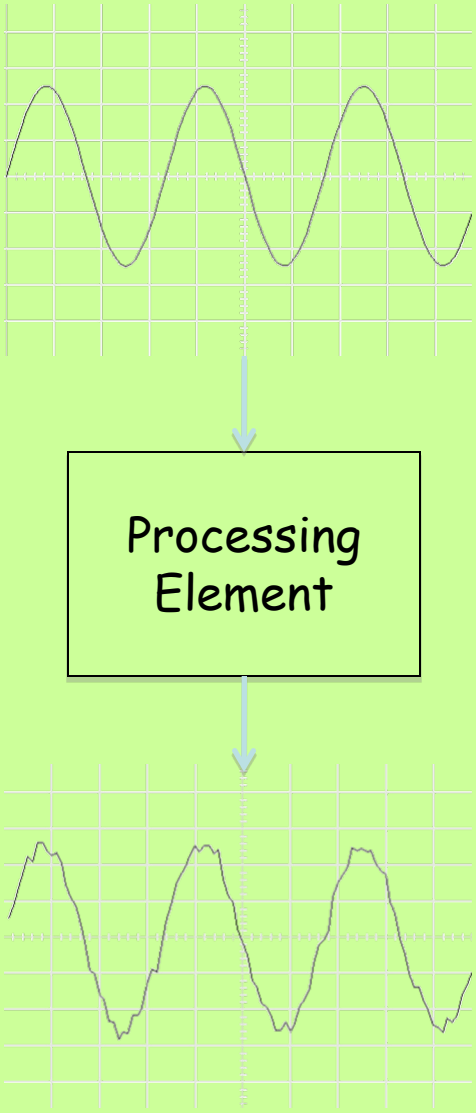
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Analog vs Digital



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The trouble with analog signaling



The real world is full of continuous-time continuous-value (aka "analog") signals created by physical processes: sound vibrations, light fields, voltages and currents, phase and amplitudes, …

But if we build processing elements to manipulate these signals we must use non-ideal components in real-world environments, so some amount of error (aka "noise") is introduced. The error comes from component tolerances, electrical phenomenon (e.g., IR and LdI/dt effects), transmission losses, thermal noise, etc. Facts of life that can't be avoided…

And the more analog processing we do, the worse it gets: signaling errors accumulate in analog systems since we can't tell from looking at signal which wiggles were there to begin with and which got added during processing.

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Analog vs Digital

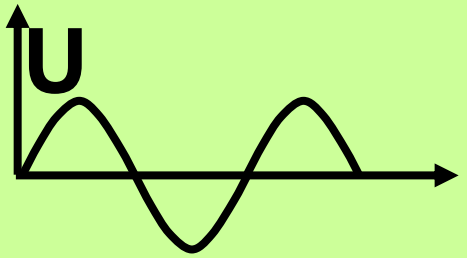
- Primitives from physical
- Noise from thermal fluctuations
- Noise accumulates

- Primitives are Boolean functions
- Noise from round-off error
- Error does not accumulate through stages

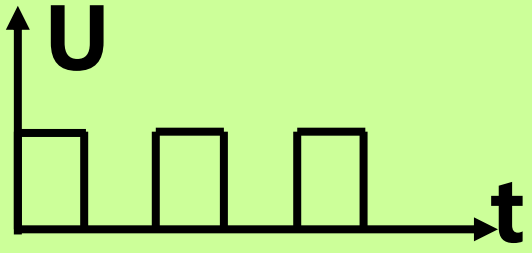
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Signals in circuit

- Analog signal
- Digital signal



Continuous time signal



Discrete in time and amplitude

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Recorder





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Camera





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Video camera





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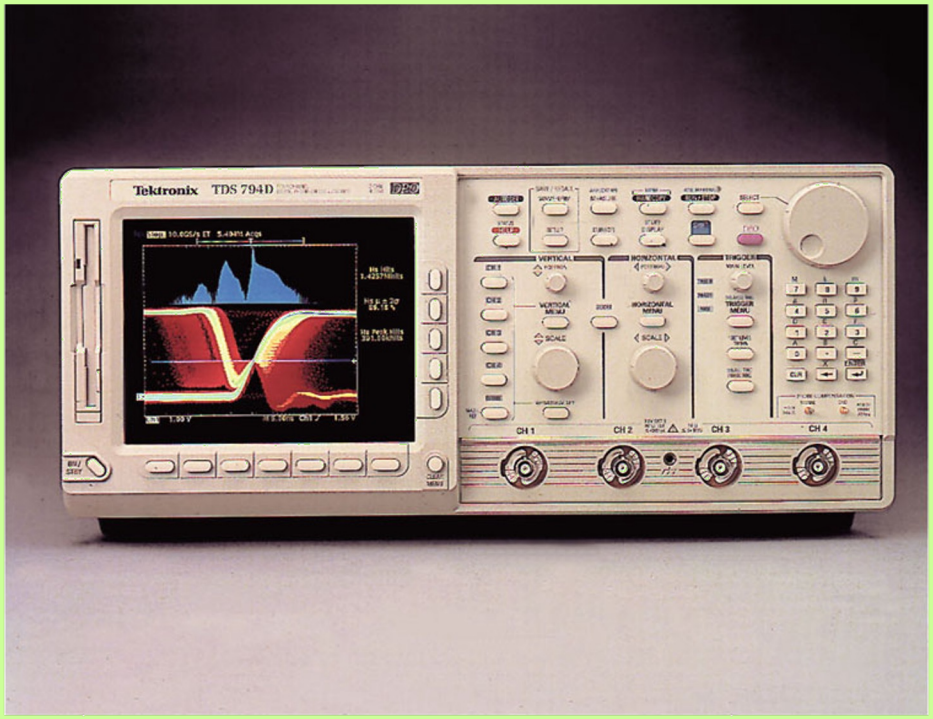
Image technology



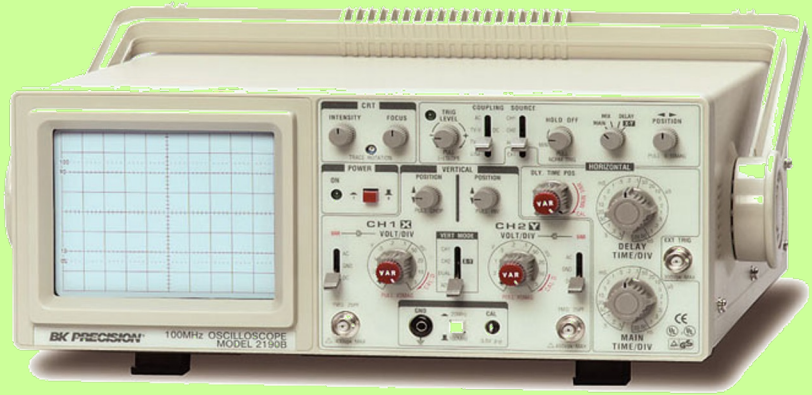




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(a) Digital oscilloscope

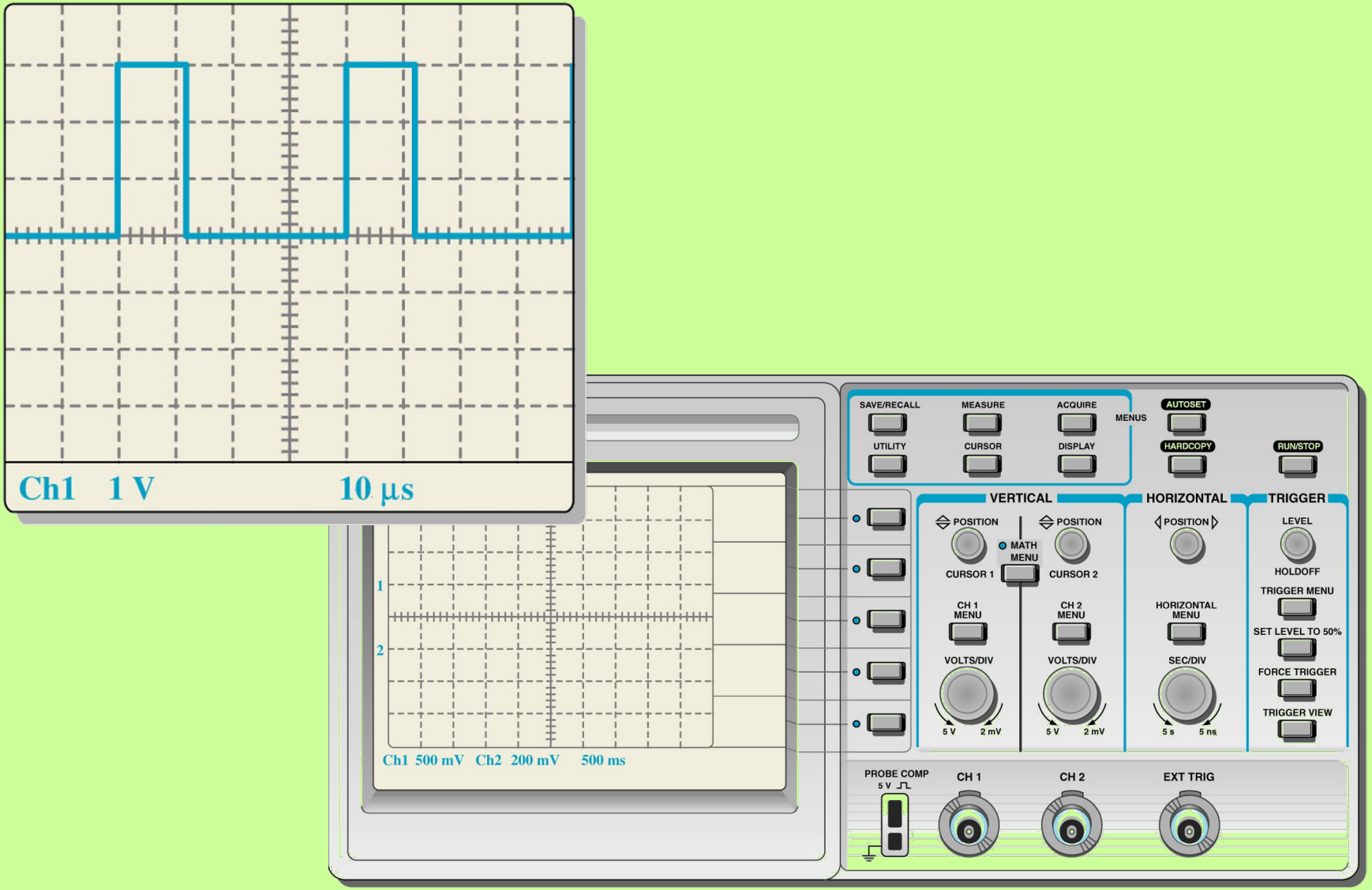


(b) Analog oscilloscope

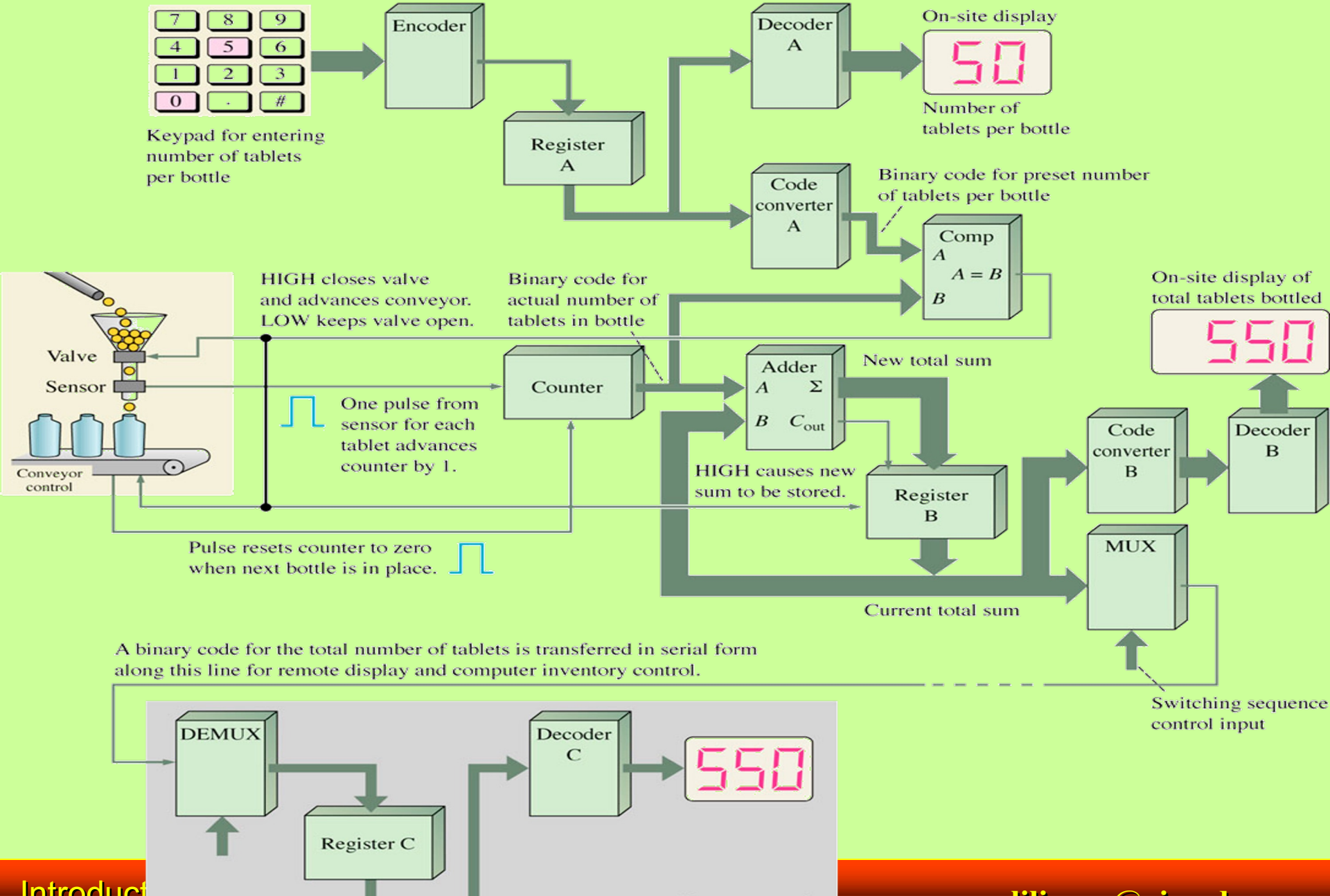


(c) Handheld oscilloscope

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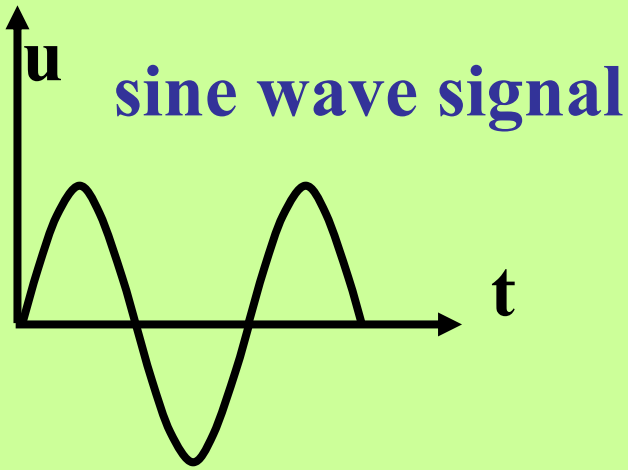


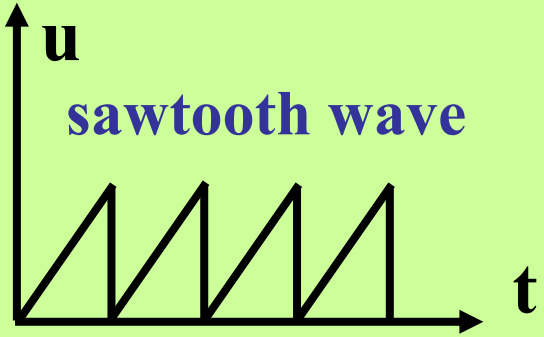
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Analog Signal And Circuit:





When studying analog circuits, we pay attention to the magnitude and phase relationship between the input and output signals of the circuit. Circuits include AC and DC amplifiers, filters, signal generators, etc.

In analog circuits, transistors generally work in an amplifying state.

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```
constant M:std_logic_vector:="101";
constant N:std_logic_vector:="010";
signal c0,c1:std_logic;
```



```
end process;
clkout<= c0 or c1;
end Behavioral;
```

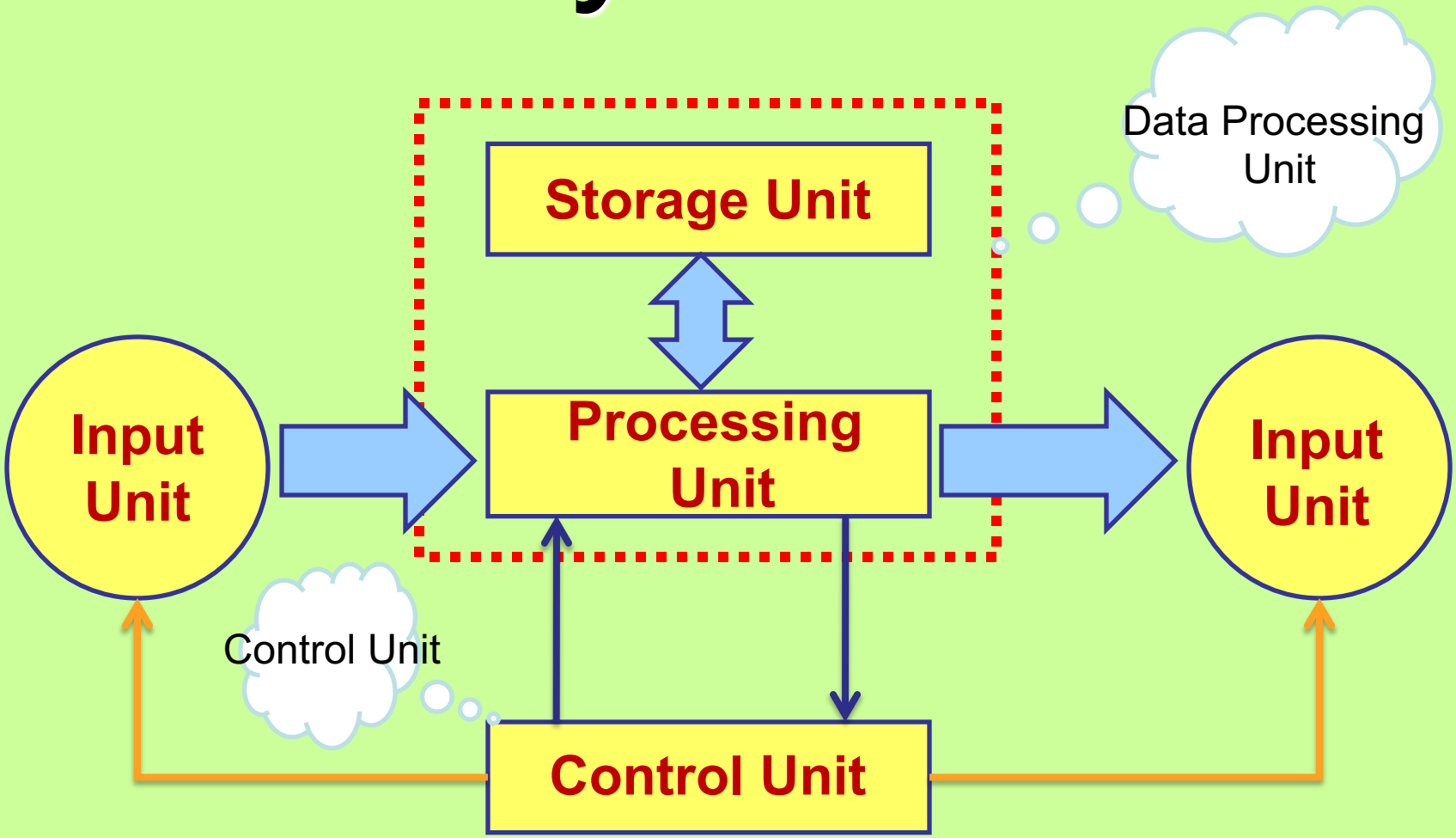

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Basic Concepts of Digital Systems

A digital system refers to an electronic system that stores, transmits, and processes digital information, and its input and output are both digital. In structure, it is divided into data processing unit and control unit.

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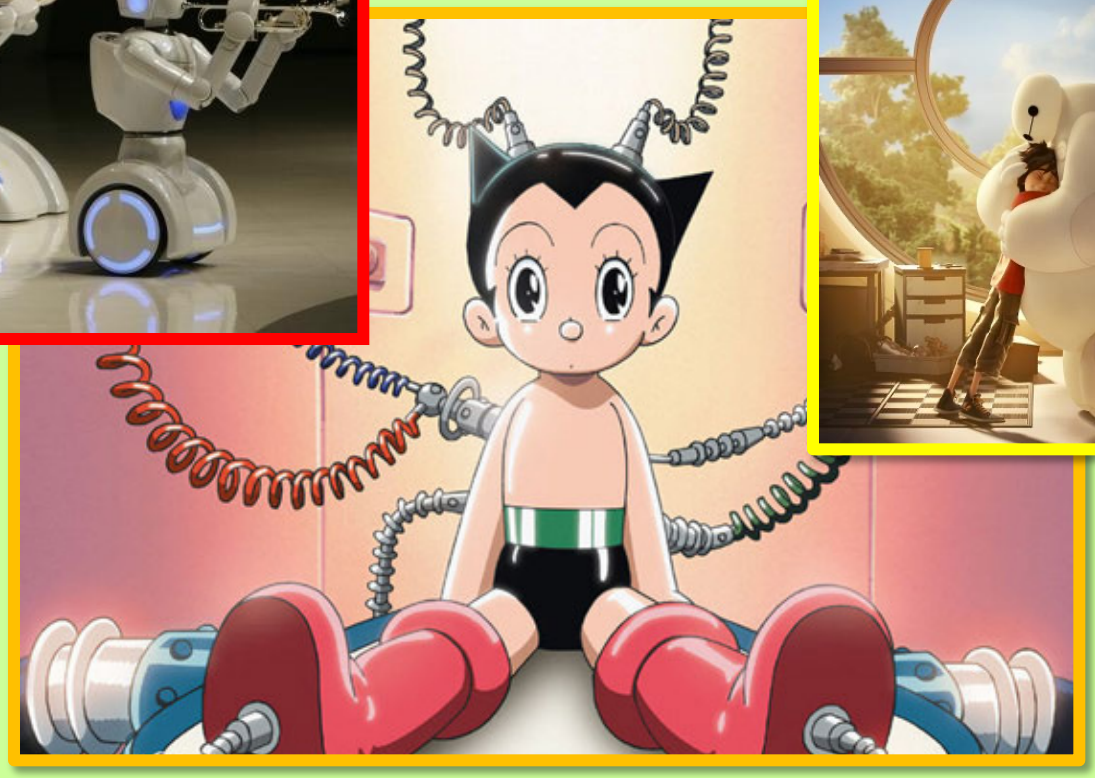
Basic Concepts of Digital Systems



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Digital Systems







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Digital Systems



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Digital Systems



Google Robot



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数字系统





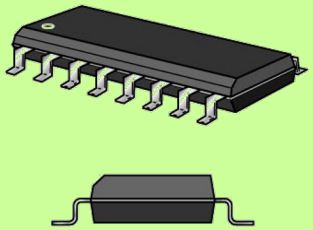






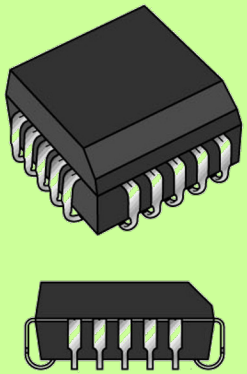


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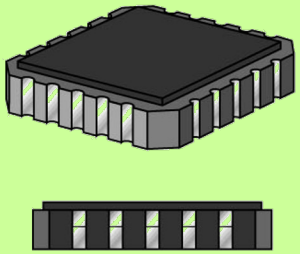
End view

(a) SOIC with "gull-wing" leads



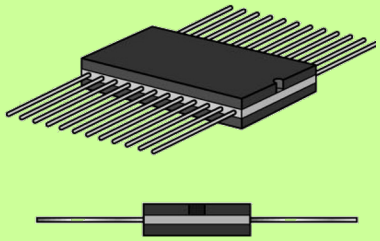
End view

(b) PLCC with J-type leads



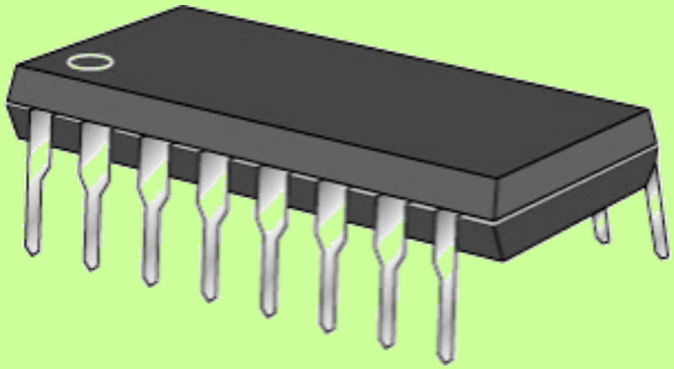
End view

(c) LCCC with no leads (contacts are part of case)

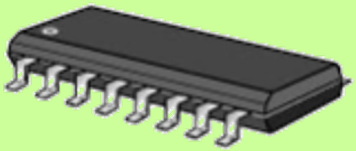


End view

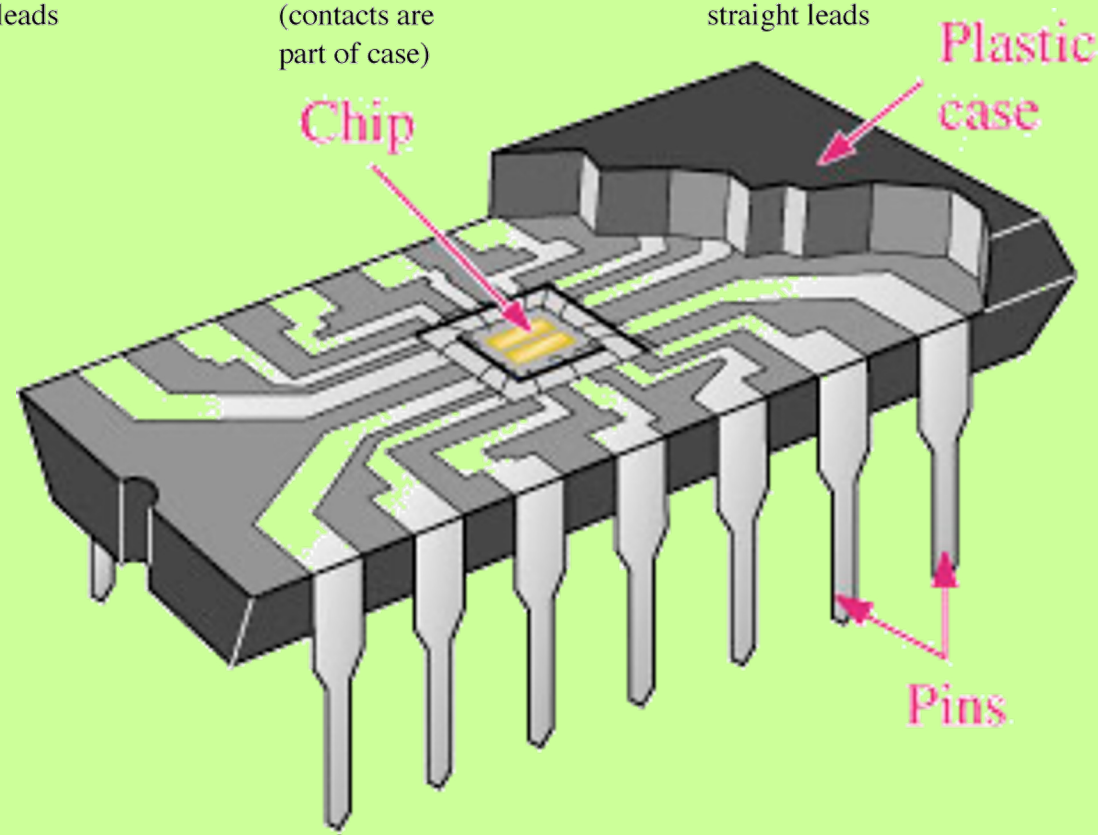
(d) Flat pack (FP) with straight leads



(a) Dual in-line package (DIP)



(b) Small-outline IC (SOIC)



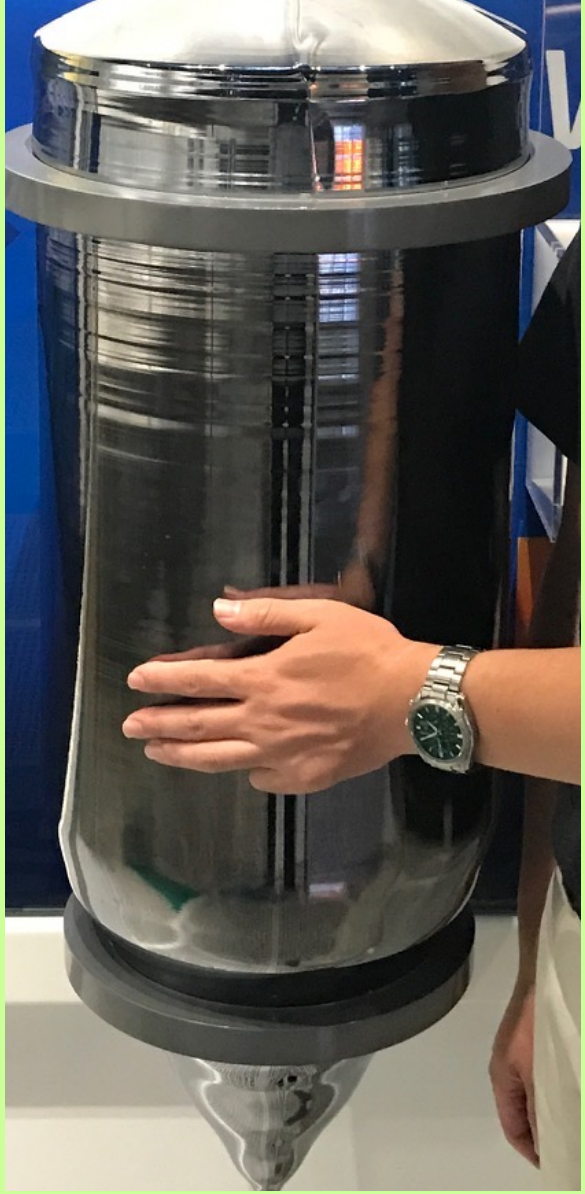
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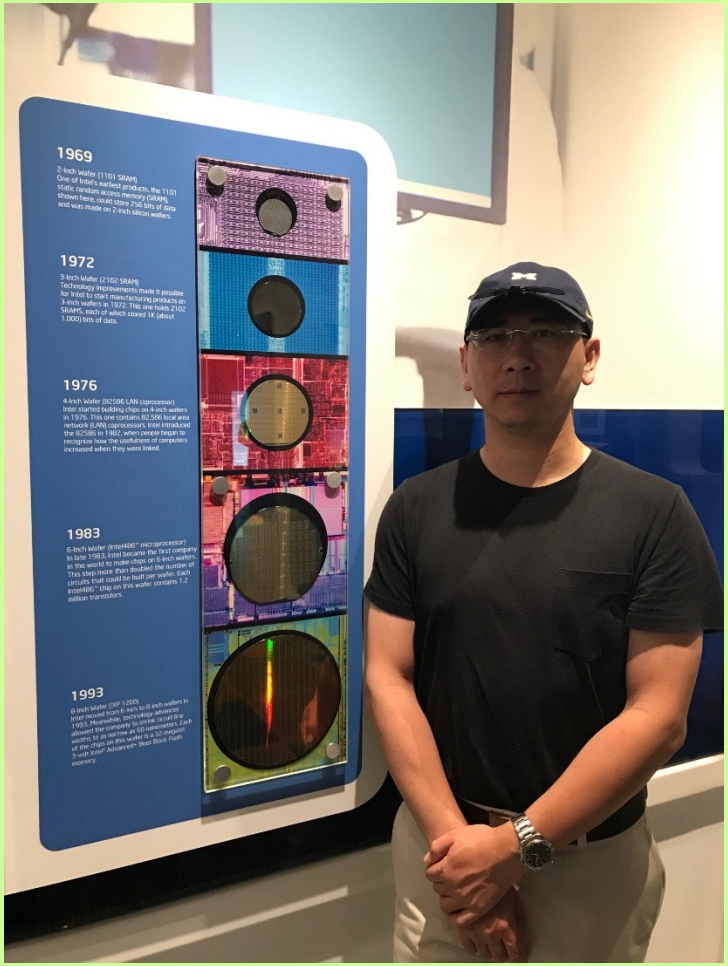




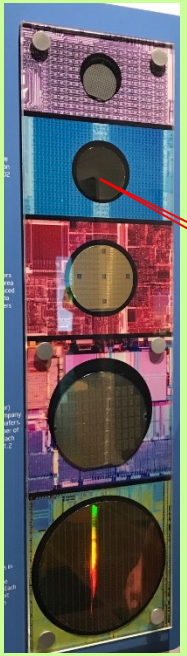


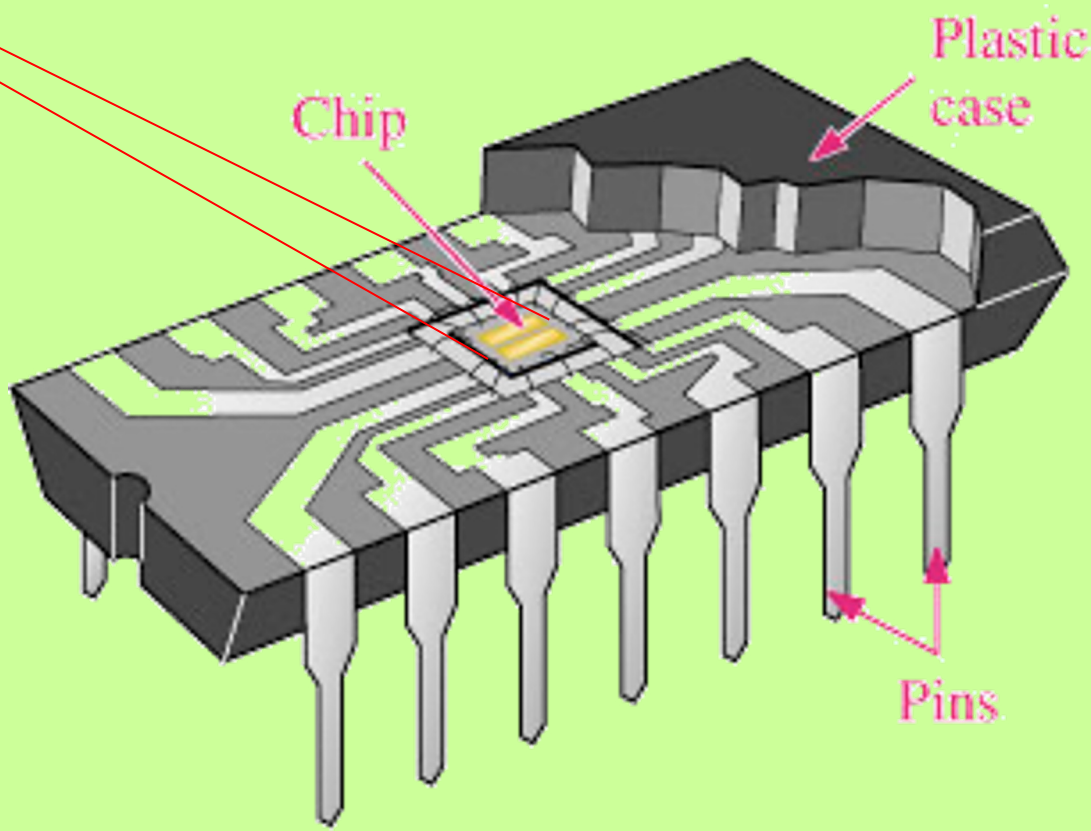
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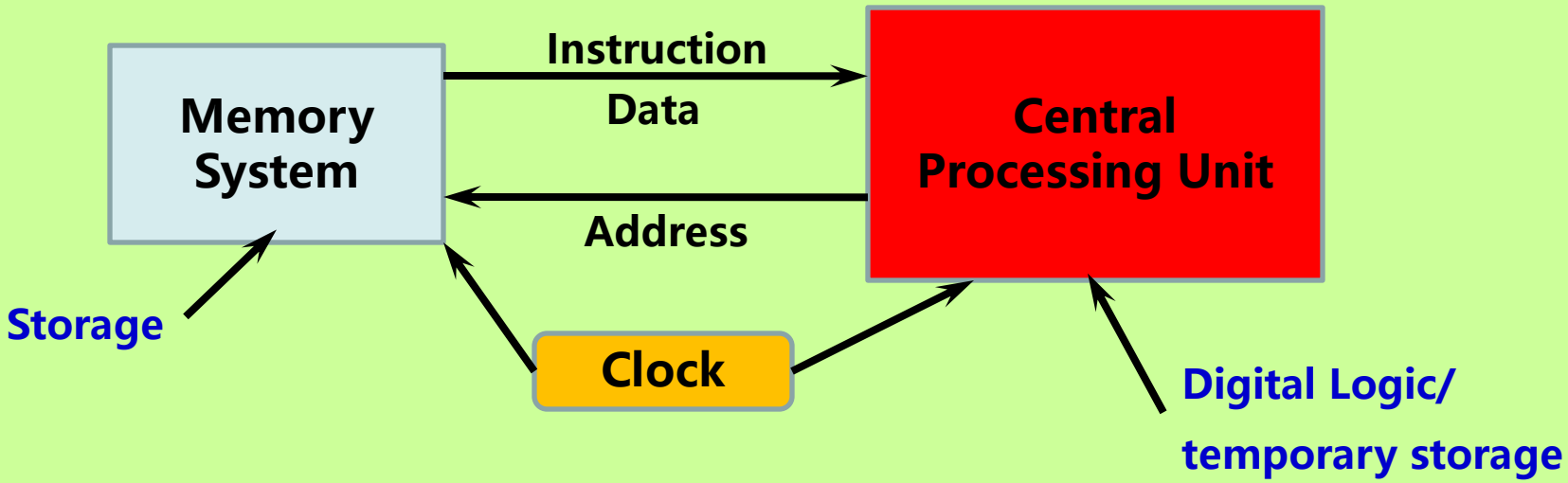
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Plastic
case
Chip
Pins

Basic Concepts of Digital Systems

Example: General Digital Computer



The design goal is to achieve optimal operating speed performance

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Basic Concepts of Digital Systems

Portable Computing Device



The design goal is to achieve lower cost of production

"Lower cost" comes at the expense of "lower running speed".

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Basic Concepts of Digital Systems

Digital Clock



The design goal is to achieve lower power consumption

"Lower power consumption" comes at the expense of "lower running speed "、 "Higher cost".

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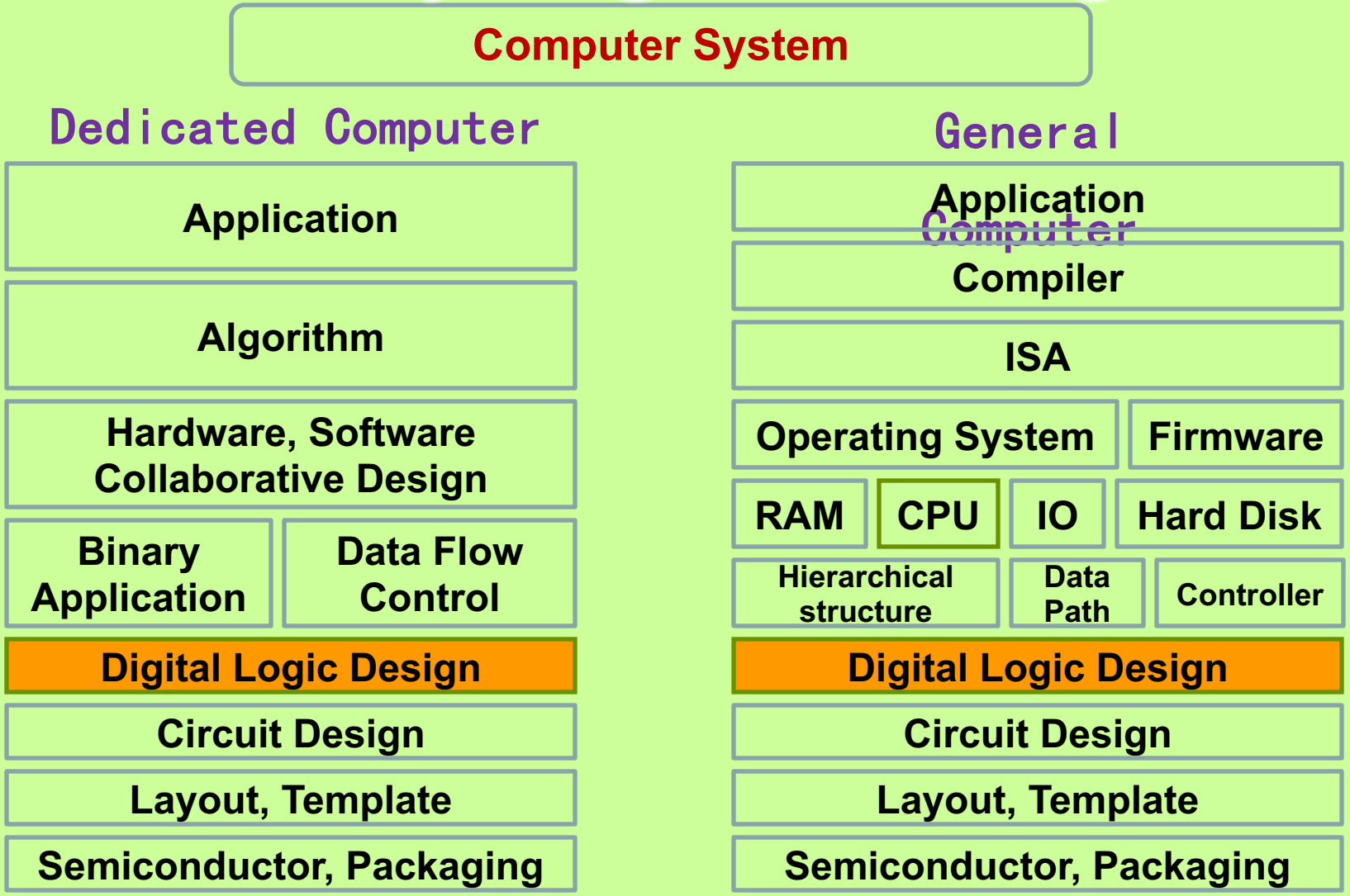
Typical Digital System



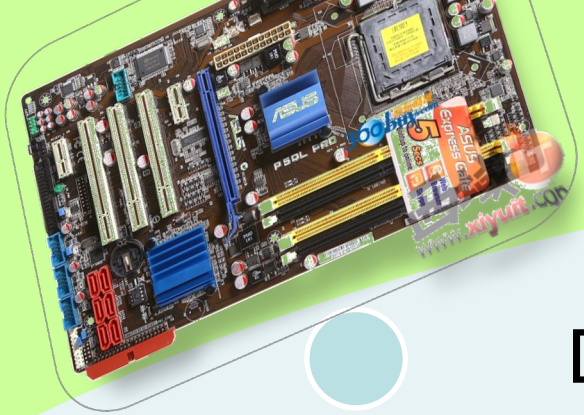


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Why Digital Design



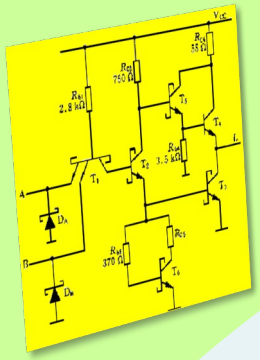
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Digital System

Circuit Board

Chip

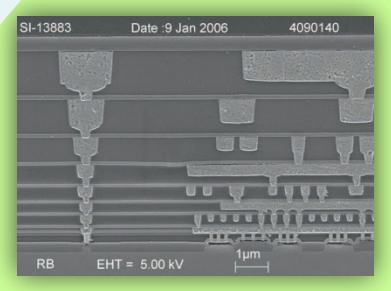






Gate Circuit

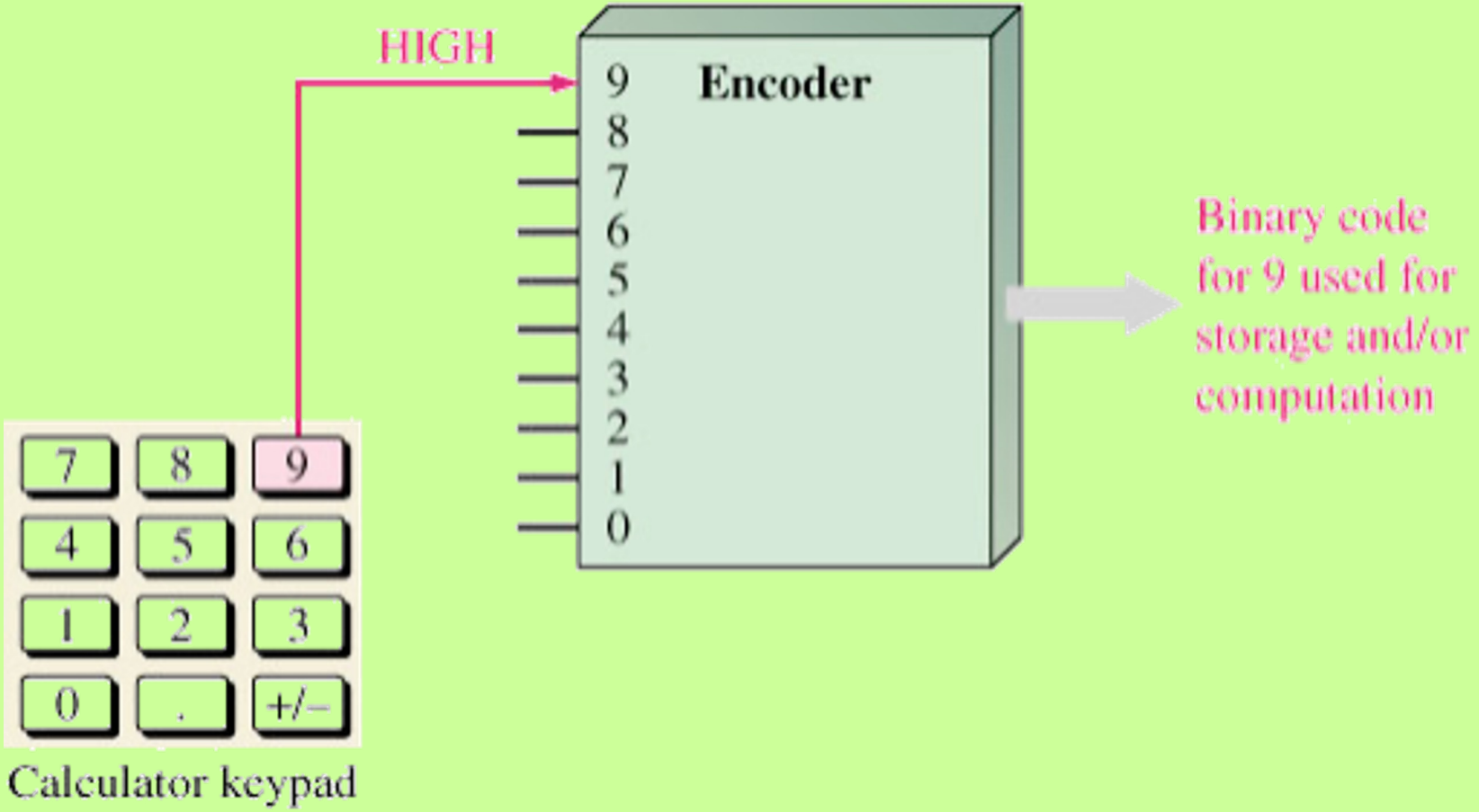
Transistor



Building Blocks of Digital System Hardware

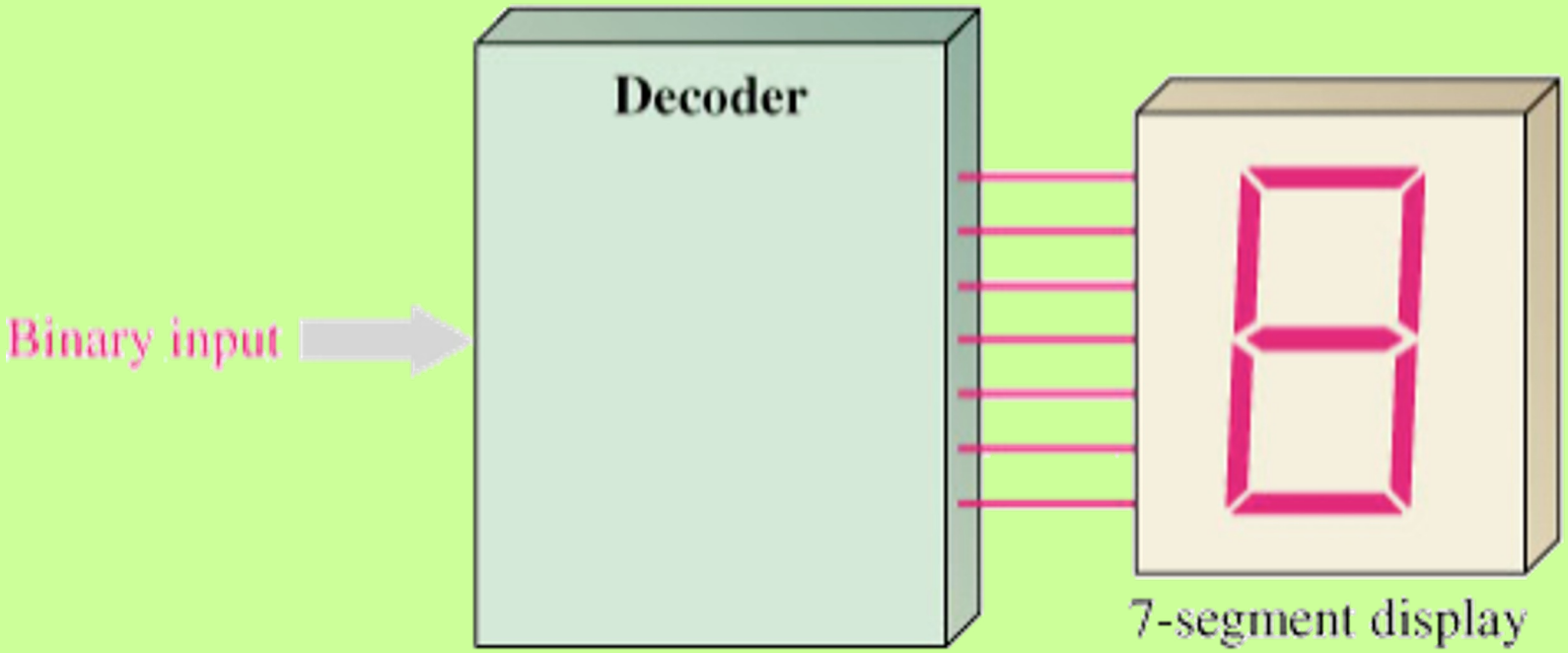
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Encoder in Counter



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Decoder in Calculator

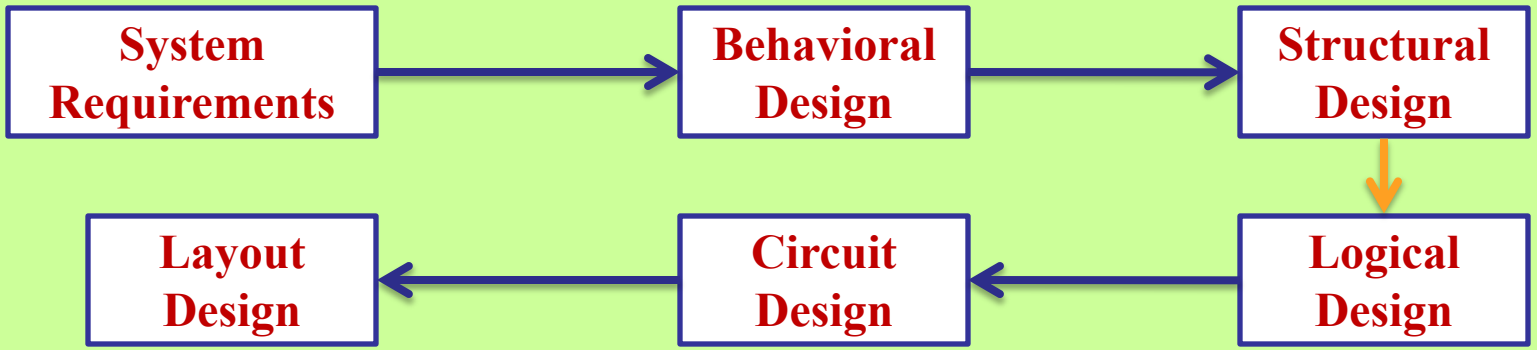


Basic Concepts of Digital Systems

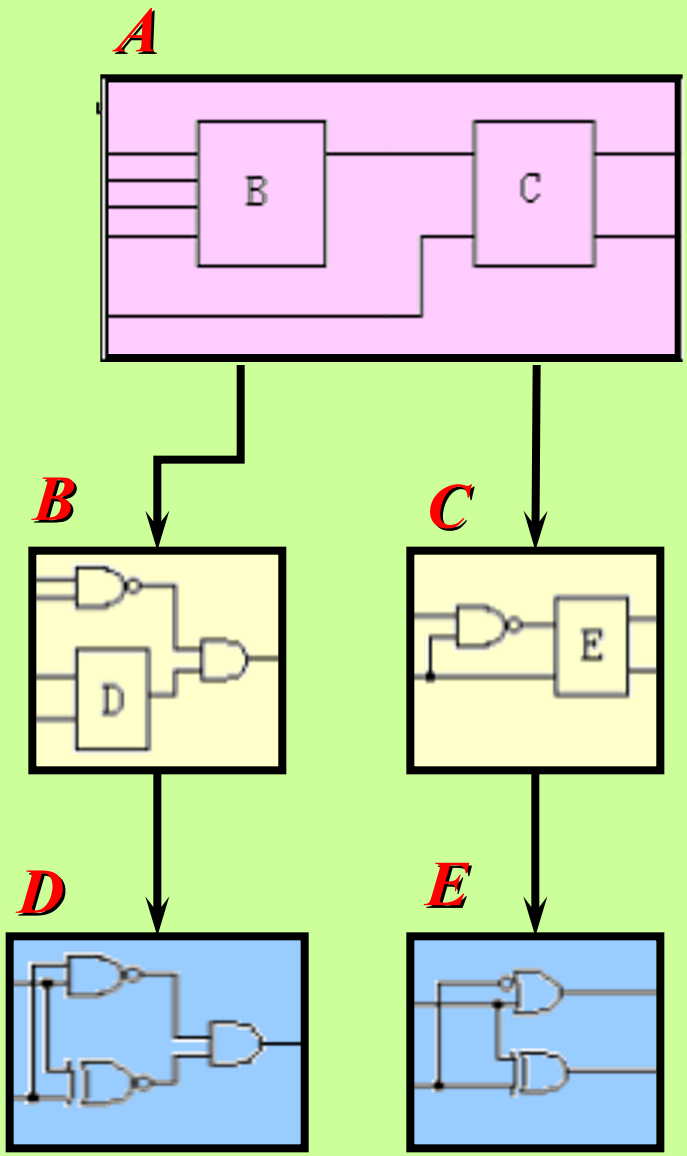
Design Method

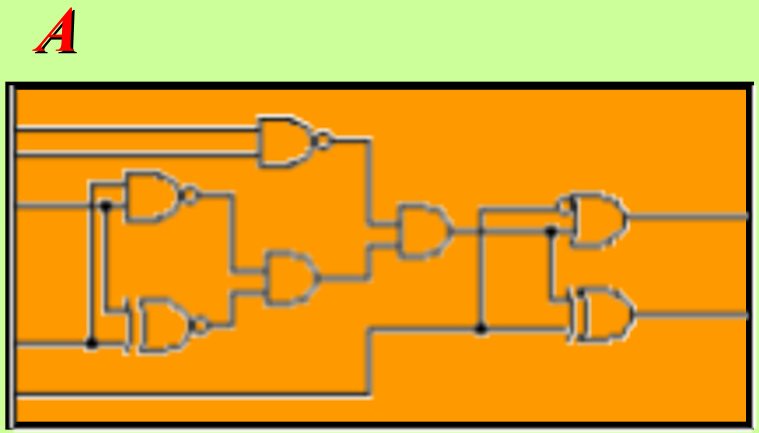
Traditional digital system design is a bottom-top design

Modern digital system design is oriented to user requirements, that it completes the corresponding description, synthesis, optimization, simulation and verification in a top-bottom way, according to the system behavior and functional requirements.



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Top-Bottom Design

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Trends in Electronic Design

Electronic products are growing rapidly, moving towards diversification, minimization of volume, and minimization of power consumption, while prices are also on the decline. An example is given of ICCAD(Integrated Circuit Computer-Aided Design) system for integrated circuit design.

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Trends in Electronic Design

Development of ICCAD:

First Generation: Late 1960s, Layout Editing and Inspection

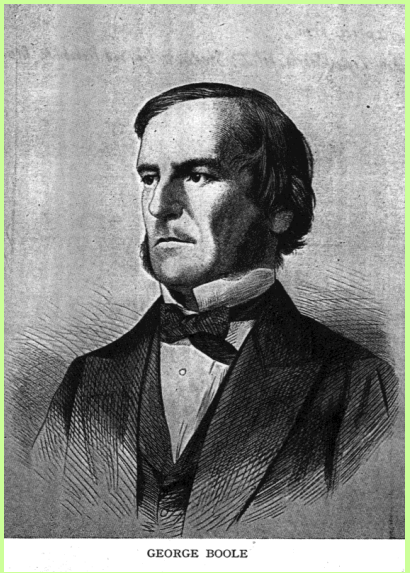
Second Generation: Early 1980s, Schematic Input, Logical Simulation Down

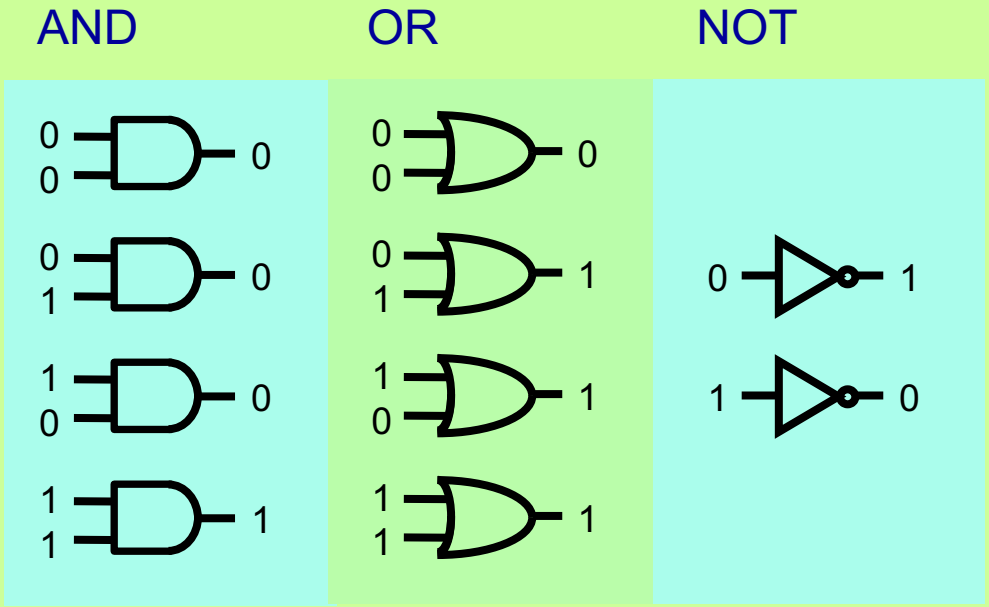
Third Generation: Down from RTL level, including behavior simulation, behavior synthesis, logic synthesis, etc.

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Logic is Math

- **To design a microprocessor is to design its logic circuit to do whatever it is intended to do**



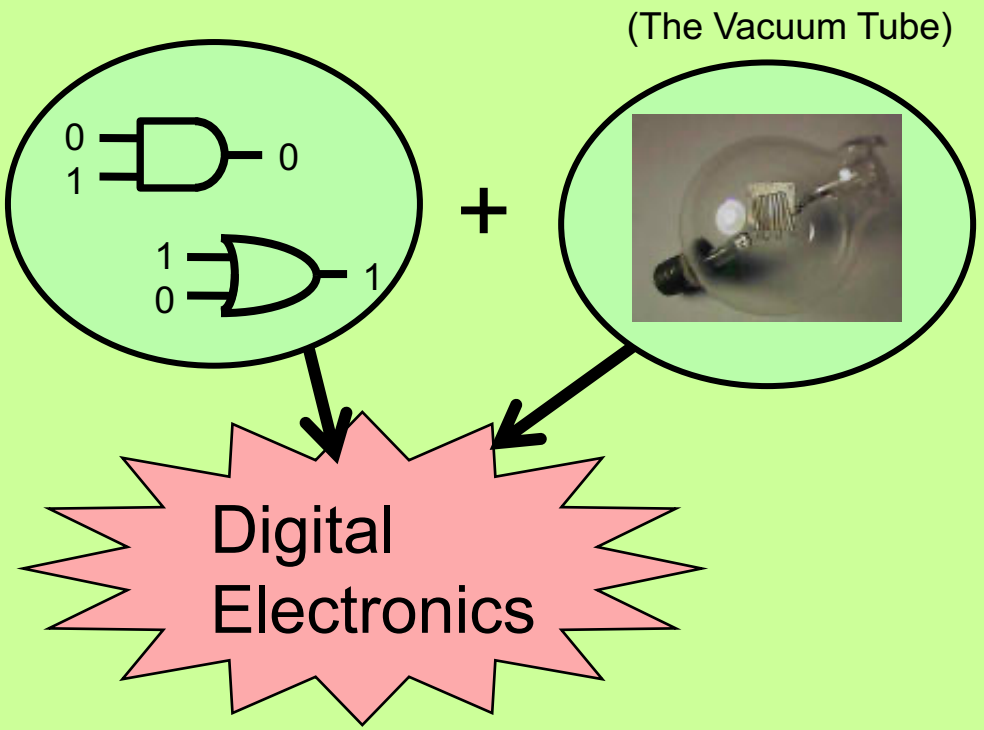


- 1854: George Boole shows that logic is math, not just philosophy!
- Boolean algebra: the mathematics of binary values

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Key Link Between Logic and Circuits

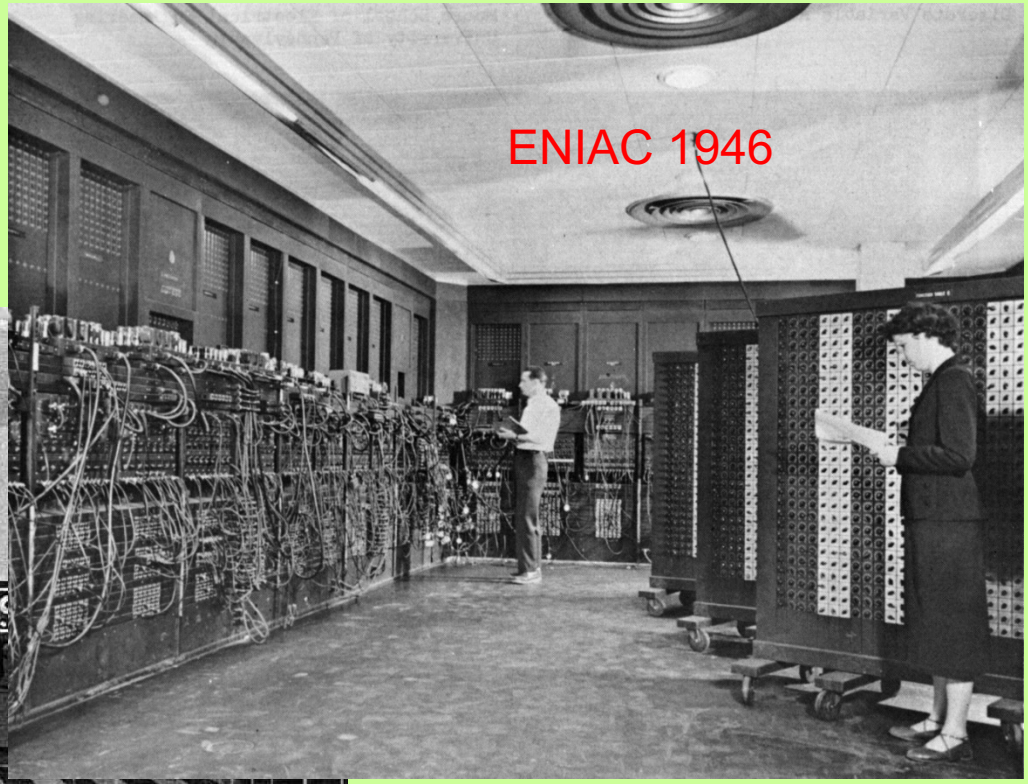
- Despite existence of relays and introduction of vacuum tube in 1906, digital electronics did not emerge for thirty years!
- Claude Shannon notices similarities between Boolean algebra and electronic telephone switches
- Shannon's 1937 MIT Master's Thesis introduces the world to binary digital electronics

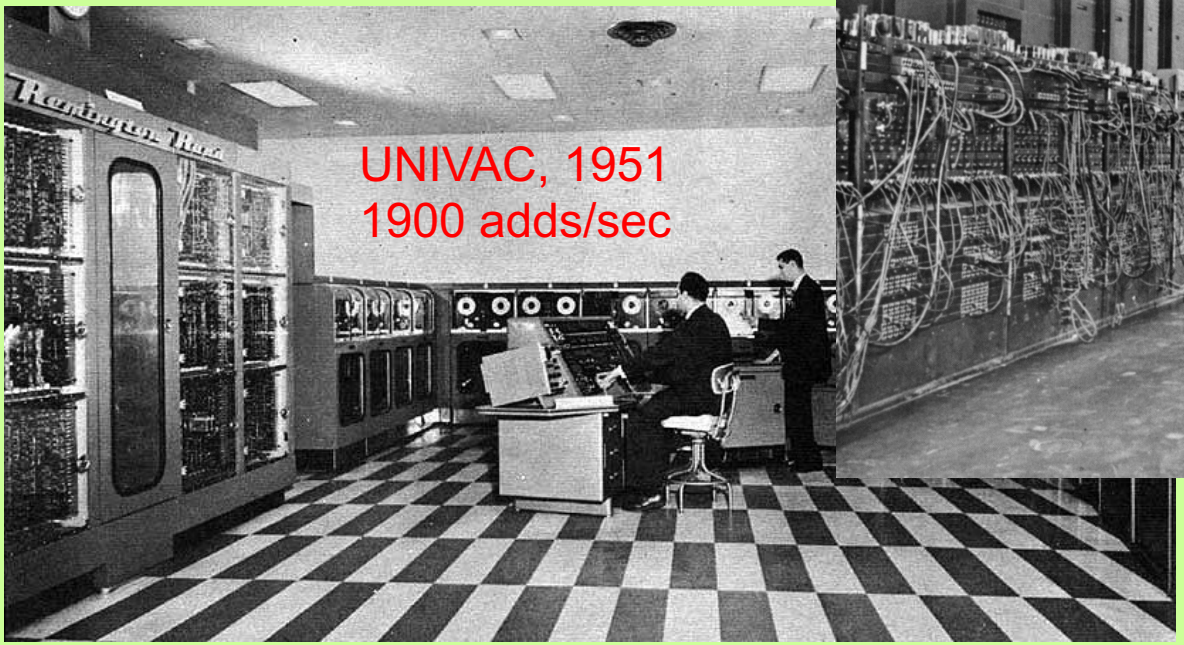


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Vacuum Tubes



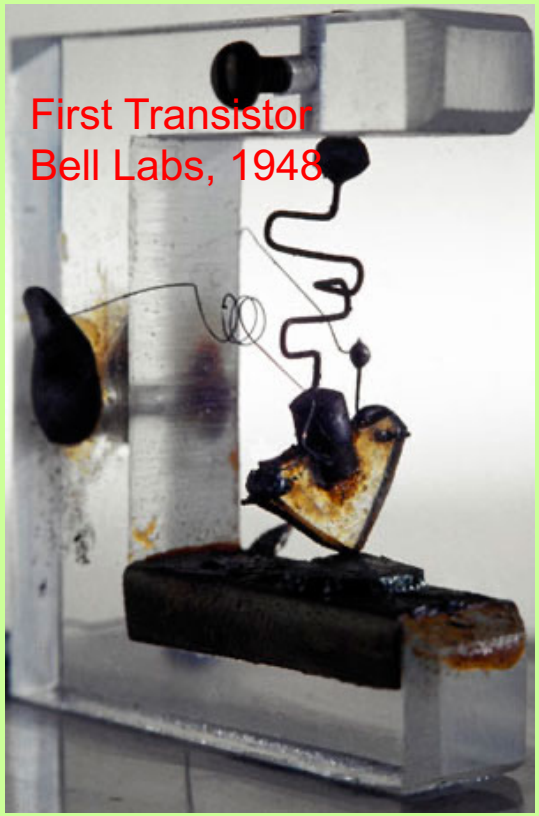




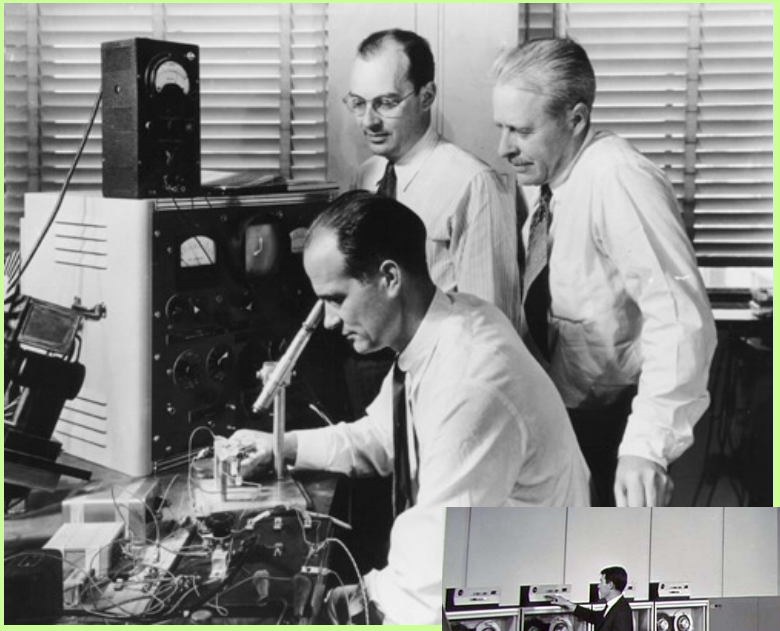
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Transistors





First Transistor
Bell Labs, 1948





IBM System/360, 1964

500,000 adds/sec

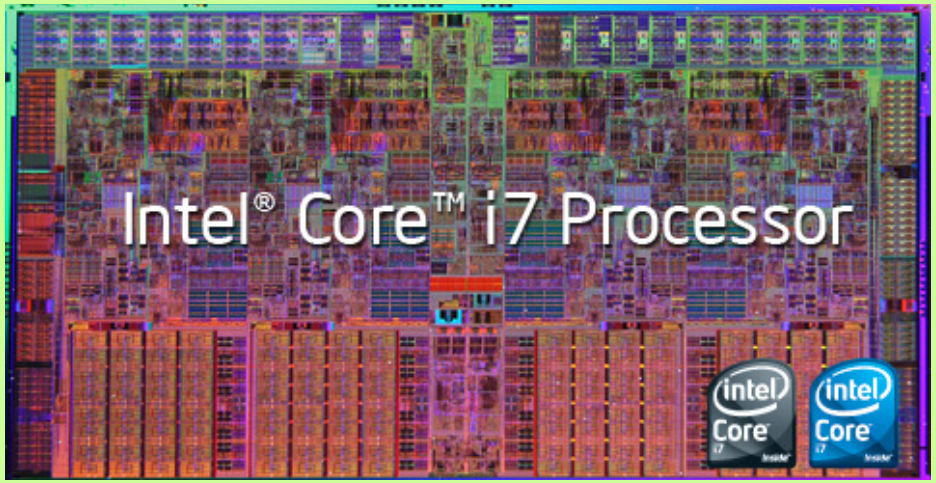
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VLSI Circuits





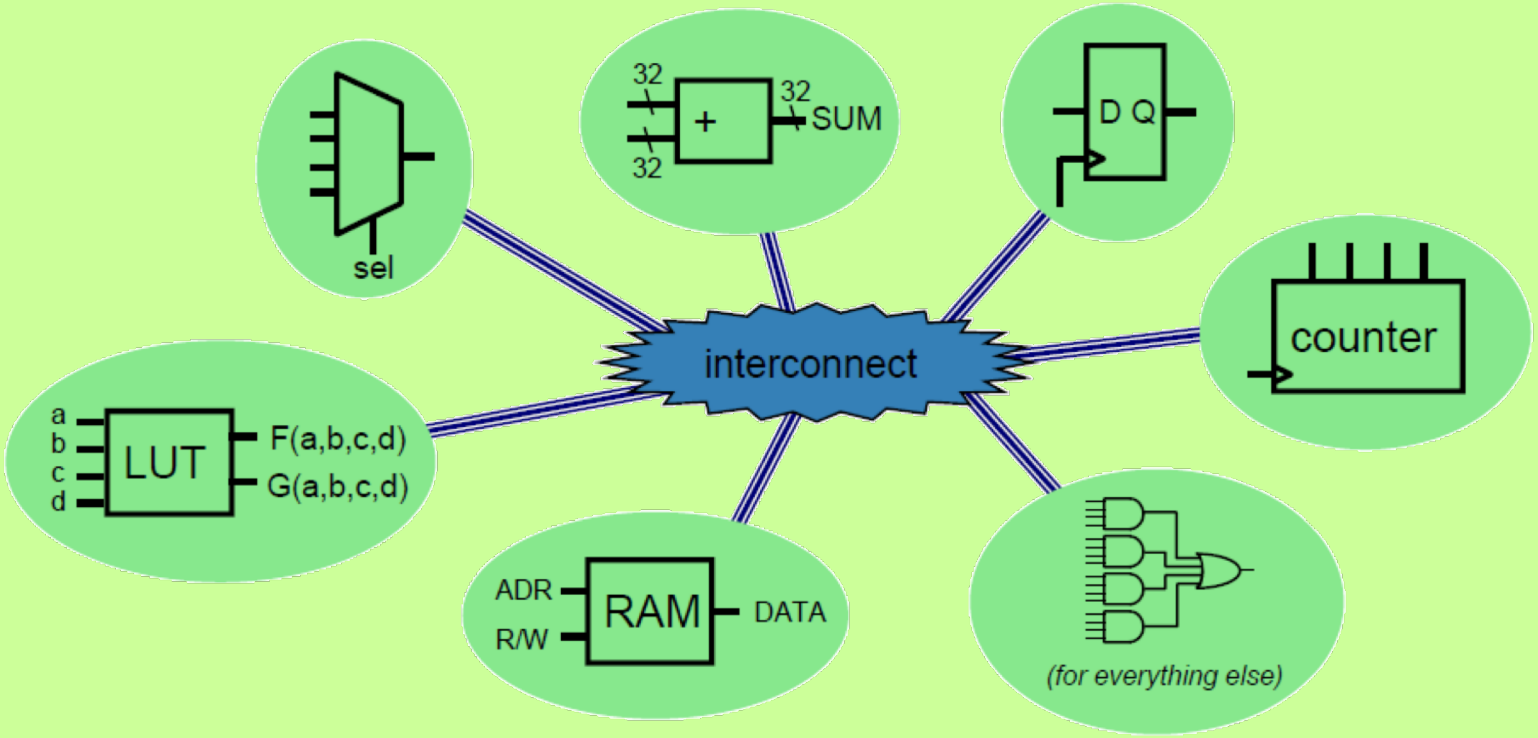




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The FPGA: A Conceptual View

- An FPGA is like an electronic breadboard that is wired together by an automated synthesis tool
- Built-in components are called macros



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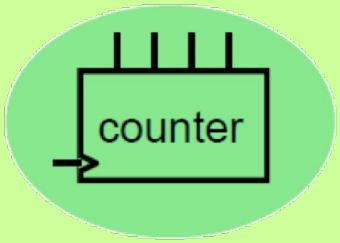
Synthesis and Mapping for FPGAs

- Infer macros: choose the FPGA macros that efficiently implement various parts of the HDL code

```
...
always @ (posedge clk)
begin
    count <= count + 1;
end
...
```

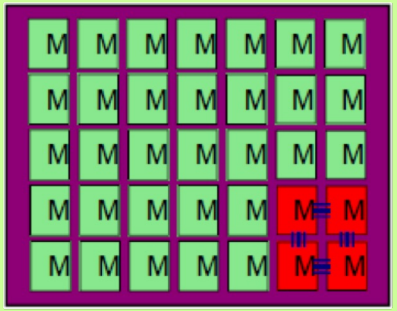
HDL Code

"This section of code looks like a counter. My FPGA has some of those..."



Inferred Macro

- Place-and-route: with area and/or speed in mind, choose the needed macros by location and route the interconnect



"This design only uses 10% of the FPGA. Let's use the macros in one corner to minimize the distance between blocks."

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FPGA

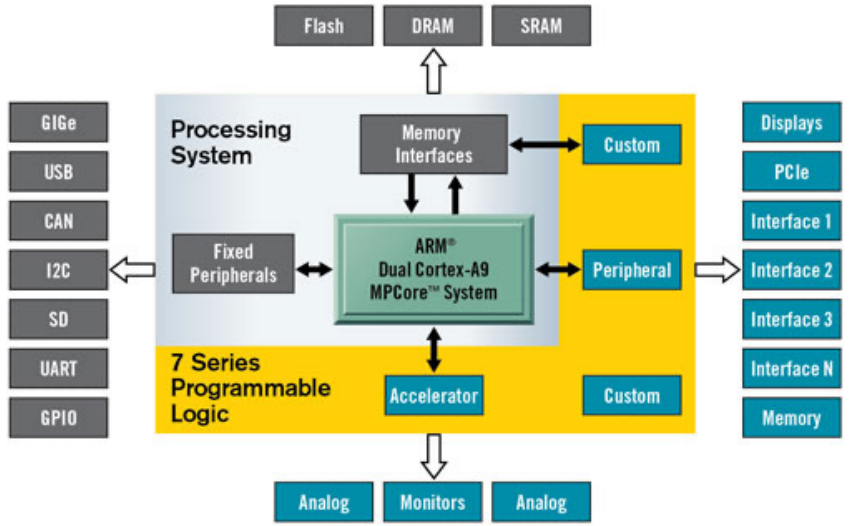
Field-Programmable Gate Arrays (FPGAs) are semiconductor devices that are based around a matrix of configurable logic blocks (CLBs) connected via programmable interconnects.

FPGAs can b
functionality

Two major br

- **Xilinx**
- **Altera**

	Spartan-6	Artix-7	Kintex-7	Virtex-7	Kintex UltraScale	Virtex UltraScale
Logic Cells					0	4,407,480
BlockRAM						132.9Mb
DSP Slices						2,880
DSP Performance (symmetric					ACs	4,268 GMACs
Transceiver Count						120
Transceiver Speed					s	32.75 Gb/s
Total Transceiver Bandwidth (					/s	5,886 Gb/s
Memory Interface (DDR3)						2,400
PCI Express® Interface					3	x8 Gen3
Analog Mixed Signal (AMS)/X						System Monitor
Configuration AES						Yes
I/O Pins						1,456
I/O Voltage					V	1.0 – 3.3V



Please refer to the device data sheets for the latest product information.

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FPGA vs DSP vs MCU vs ASIC

- **Digital signal processors (DSPs) hard-wire the basic functions of many signal-processing algorithms. There are many high sampling rate applications that an FPGA does easily, while the DSP could not. Equally, there are many complex software problems that the FPGA cannot address.**

- **Microcontrollers (MCUs), like an Arduino, the chip is already designed for you. You simply write some software, usually in C or C++, and compile it to a hex file that you load onto the microcontroller. FPGAs are different. You are the one designing the circuit. You can configure an FPGA to be something as simple as an and gate, or something as complex as a multi-core processor. To create your design, you write some HDL (Hardware Description Language).**

- **Application Specific Integrated Circuits (ASICs), which are custom manufactured for specific design tasks.**

With FPGAs you have control over the **hardware**

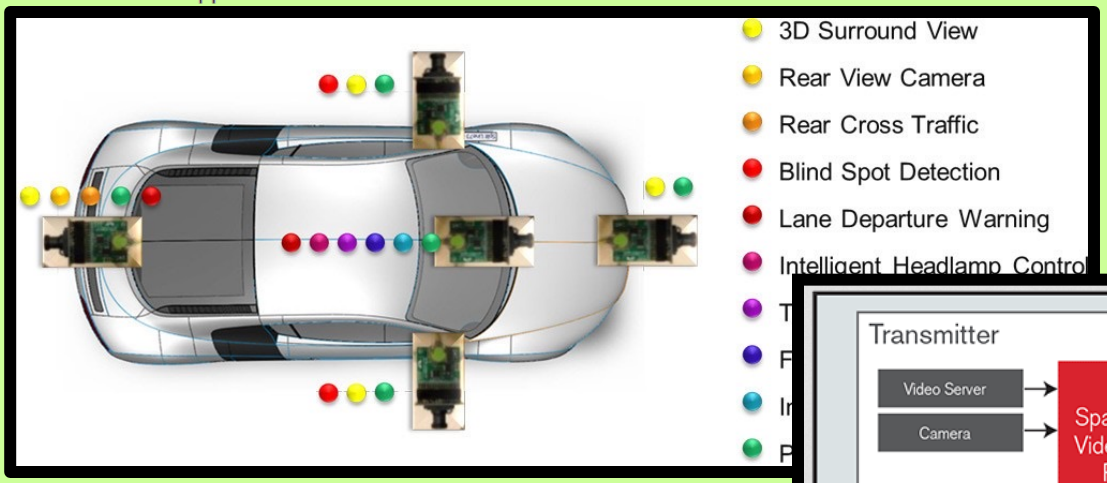
Hardware Description Language

- A hardware description language (HDL) is a specialized computer language used to program the structure, design and operation of electronic circuits, and most commonly, digital logic circuits.
- VHDL & Verilog
  - VHDL (Very high speed integrated circuit Hardware Description Language) became IEEE standard 1076 in 1987.
  - The Verilog hardware description language has been used far longer than VHDL and has been used extensively since it was launched by Gateway in 1983.
  - Cadence bought Gateway in 1989 and opened Verilog to the public domain in 1990.
  - Verilog became IEEE standard 1364 in December 1995.

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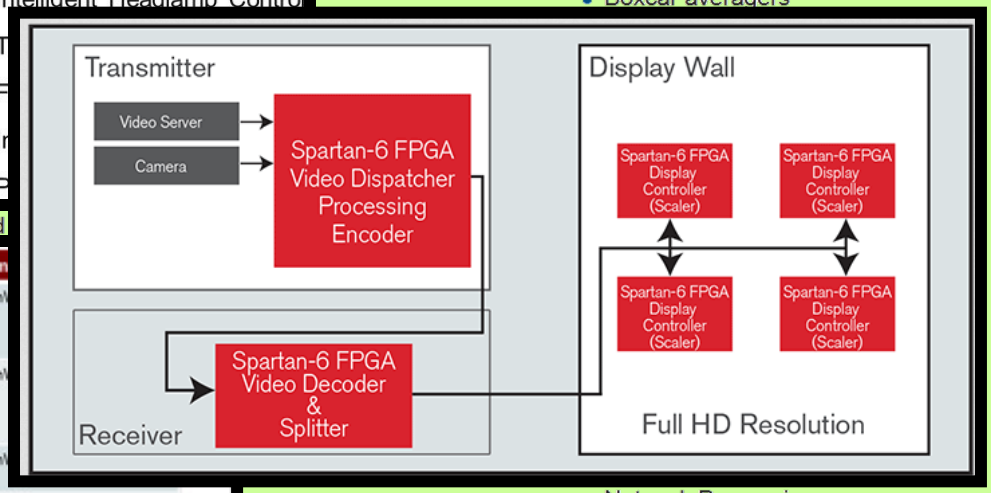
Applications

Common FPGA Applications:



- 3D Surround View
- Rear View Camera
- Rear Cross Traffic
- Blind Spot Detection
- Lane Departure Warning
- Intelligent Headlamp Control

- Digital Signal Processing (DSP)
- Load





Equipment	Configuration	Target Device	Estim
Femtocell	• 1Tx, 1Rx • 1x20MHz LTE DUC/DDC	Zynq -7010	307m
Picocell	• 1Tx, 1Rx • 1x20MHz LTE DFE	Zynq -7020	708m
Microcell	• 2Tx, 2Rx • 1x20MHz LTE DFE	Artix -7 7A100T	710m
		Kintex -7 7K70T	1.036W
		Zynq -7020	1.082W
Macrocell/ RRH	• 4Tx, 4Rx • 1x20MHz LTE DFE	Kintex -7 7K160T	2.181W
		Zynq -7030	2.38W
AAS	Centralised Module • 4Tx, 4Rx • 1x20MHz LTE DUC/DDC	Kintex -7 7K325T	4.7W
	Radio Module • 4TX, 4RX • CFR, DPD	Kintex-7 7K160T	2.22W

*w/o I/O power

- MRI
- X-ray
- PET
- Surgical Systems

Scientific Instruments

- Lock-in amplifiers
- Boxcar averagers

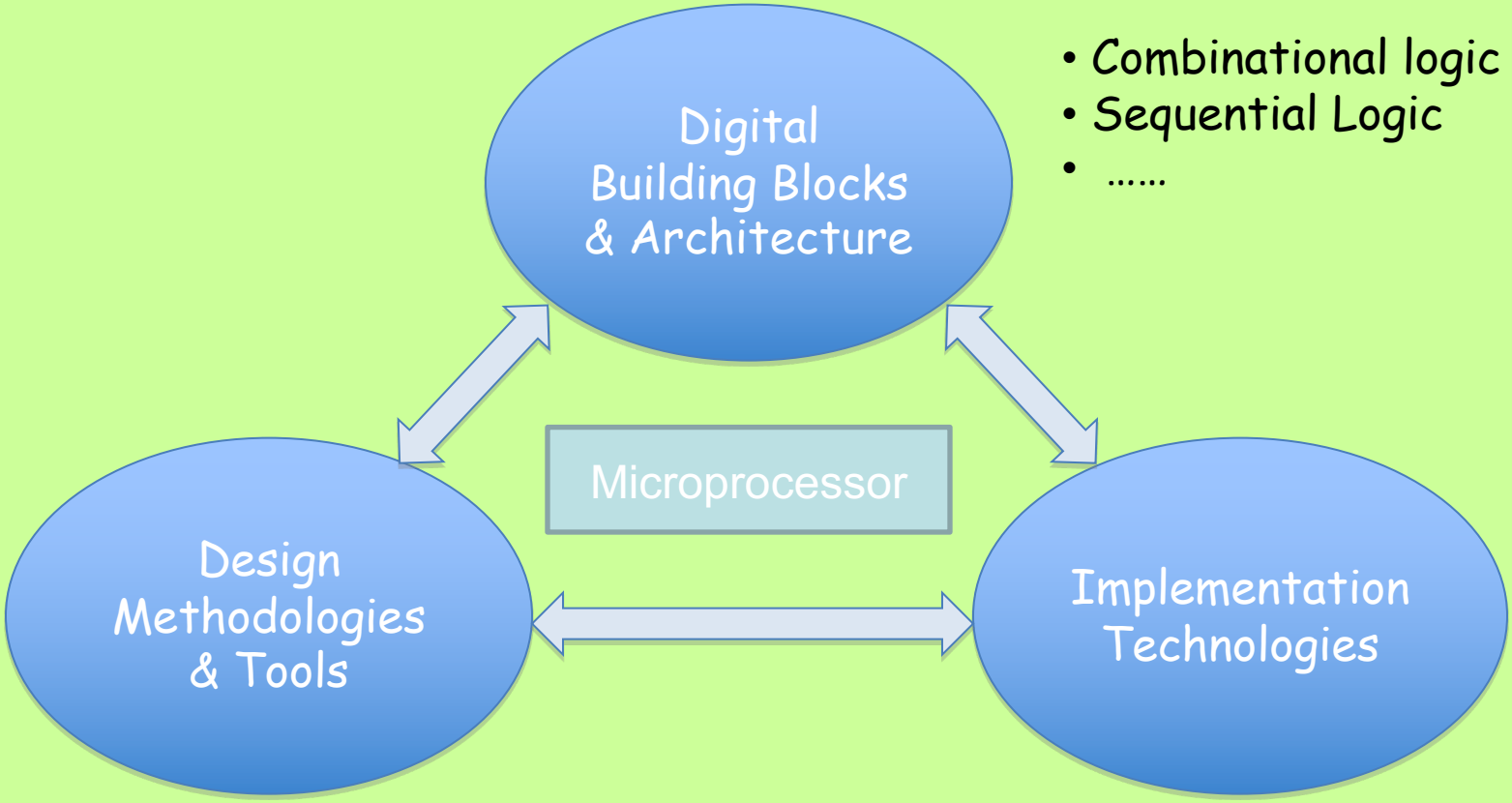
- Network Processing
- Connectivity Interfaces

Wireless Communications

- Baseband
- Connectivity Interfaces
- Mobile Backhaul
- Radio

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Topics in this course



- Combinational logic
- Sequential Logic
- ……

- HDL: Verilog
- ……

- FPGAs
- ……

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Goal

- Learn about digital logic and microprocessor design
- Understand the theory of digital logic circuits
- Be able to design and implement dedicated microcontrollers and general-purposed microprocessors
- How to interface the microprocessors with the real world

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Text/Reference Book

1. Enoch Hwang, Digital Logic and Microprocessor Design with VHDL, http://cec.shfc.edu.cn/download/2c91bf1b-480e-4a62-95f6-a7d1c88a0a8b.pdf
2. John F. Wakerly, Digital Design, (课程组 在学在浙大平台 提供下载)
3. David Harris, Sarah Harris,数字设计和计算机体系结构(英文版·第2版)2013-12
4. M. Morris Mano and Michael D. Ciletti, Digital Design, 5th edition, Pearson (ISBN-13: 9780132774208), 2013.
5. 阎石，数字电子技术基础，高等教育出版社.
6. D. E. Thomas, P. R. Moorby, The Verilog Hardware Description Language, 5th ed., Kluwer Academic Publishers (ISBN 1402070896), June 2002. http://link.springer.com.library.sutd.edu.sg:2048/book/10.1007/978-0-387-85344-4
7. S. Palnitkar, Verilog HDL, 2nd ed., Prentice Hall (ISBN 0130449113), 2003. http://proquestcombo.safaribooksonline.com.library.sutd.edu.sg:2048/book/electrical-engineering/semiconductor-technology/0130449113

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Course Evaluation

- **Project, 45%**
  - **Project Proposal: 5%**
  - **Final Project Demo: 30%**
  - **Final Project Report: 10%**
    (group report + individual component)
- **Homework + Quiz + Exam, 10%+5%+20%**
- **Lab, 20%**

Project (1) - Team

- The students will form a team with 3 students.
  - Estimated 49 students.
  - 17 teams in total.
  - Please start to form team from today.
- Prepare a project proposal and make a presentation around April 29 (estimated).
  - 10 minutes per team.
  - Explain the design process (e.g. from brain storming to final idea).
  - Provide clear objectives.
  - State clearly the task of each individual team member.
  - Submit finalized project proposal on May 6.

Project (2) - Budget

- Budget: No budget limits. No cash advancement.
- Receipt must submit before May 25, all claims subject to ZJU Finance guidelines, e.g. only parts and materials related to the project are claimable, receipts must be within 3 months, must be original receipt, transportation cost is not!
- Open source code is ok, but should not be a major part of the project, and need to declare and cite the source.
- Contributions from each team member must be stated clearly.

Project (3) – Demo and Report

- A final project demo and presentation on 16, 19, 21 and 23 June (or other day)
  - 20 minutes per team.
  - Will be video recorded.
  - Project will be graded based on their creativity, functionality and robustness of the design, difficulty of the project etc.
  - One or more projects will be selected for best design project and will be awarded (TBD).

- Final project report by July 6.
  - Acknowledge the use of open source code.
  - Contributions from individual team member must be clearly stated.

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Project (4) - Theme

- **Open-ended.**
- **Must emphasize digital concepts, but inclusion of analog interfaces (e.g., data converters, sensors or motors) common and often desirable.**

- **2023 Theme: The Asian Games**

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Binary Numbers

Table 1.1
Powers of Two

n	2^n	n	2^n	n	2^n
0	1	8	256	16	65,536
1	2	9	512	17	131,072
2	4	10	1,024 (1K)	18	262,144
3	8	11	2,048	19	524,288
4	16	12	4,096 (4K)	20	1,048,576 (1M)
5	32	13	8,192	21	2,097,152
6	64	14	16,384	22	4,194,304
7	128	15	32,768	23	8,388,608

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0 – 99 : how many bits? 7 bits

0 – 9999: how many bits? 14 bits

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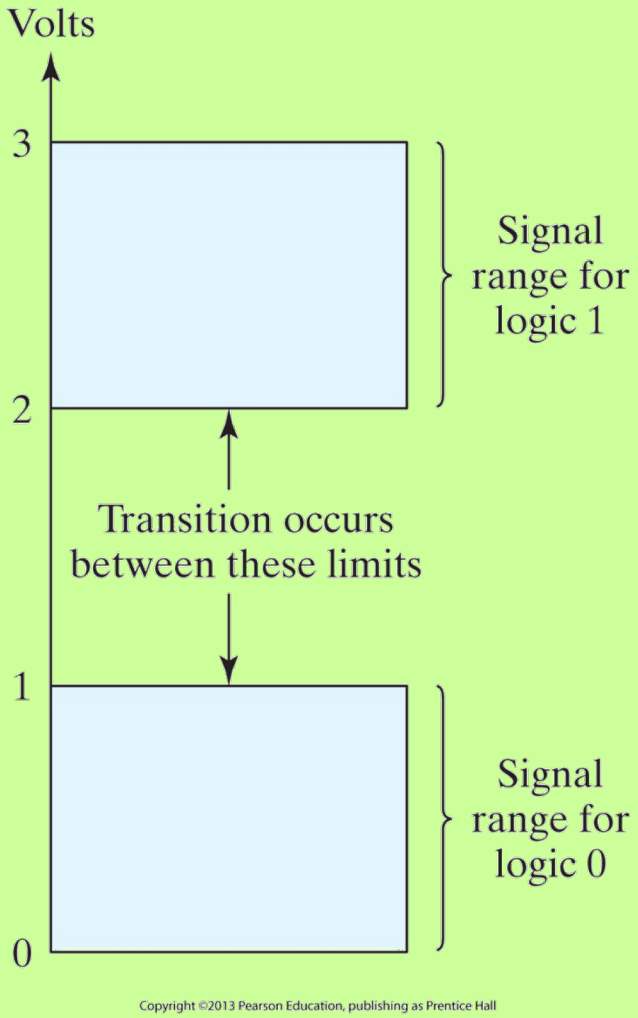
Numbers with Diff Bases

Table 1.2
Numbers with Different Bases

Decimal (base 10)	Binary (base 2)
00	0000
01	0001
02	0010
03	0011
04	0100
05	0101
06	0110
07	0111
08	1000
09	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

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Using voltages to encode binary values



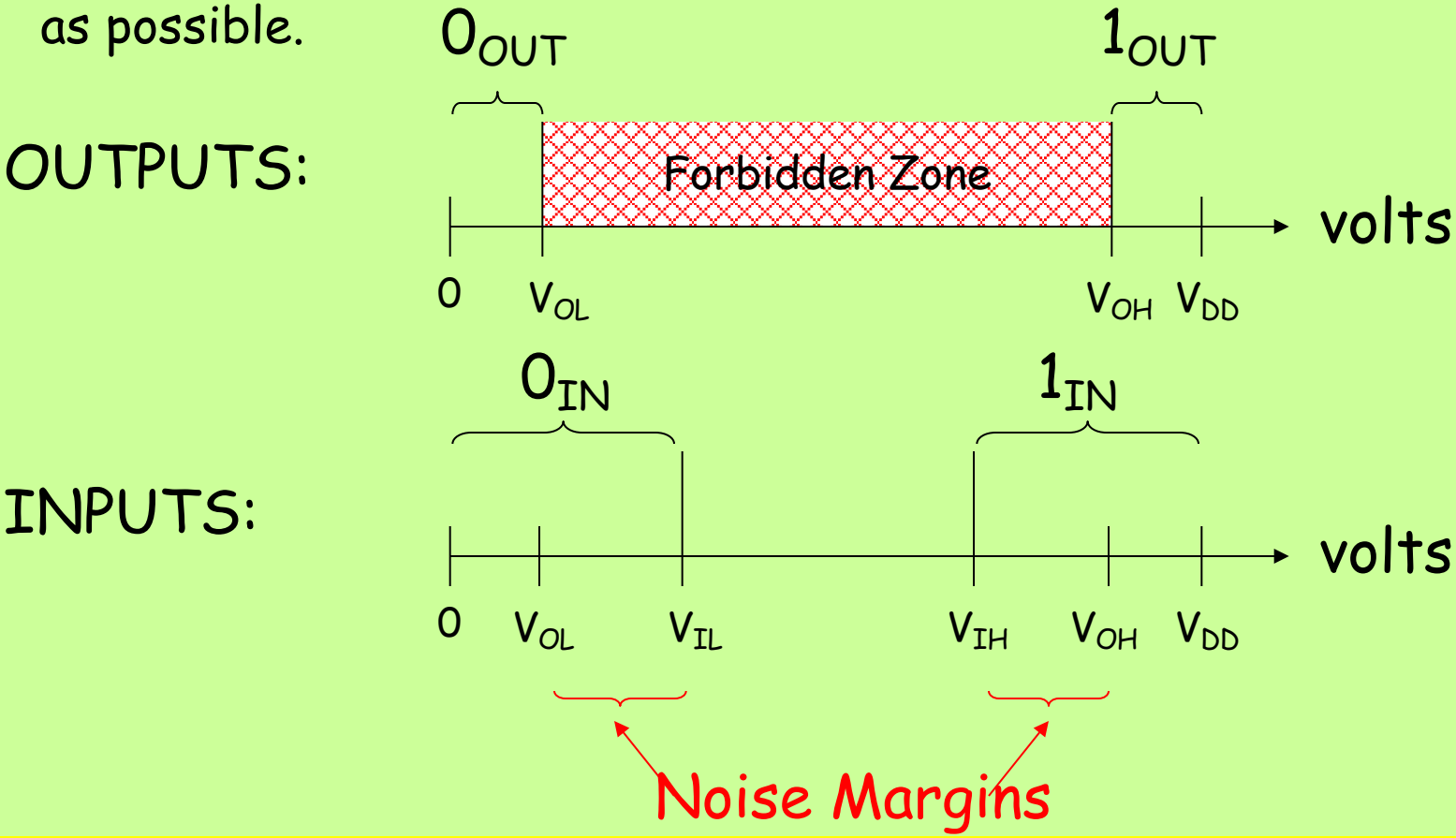
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Using analog to represent digital

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Using voltages to encode binary values

We'll keep things simple by designing our processing elements to use voltages to encode binary values (0 or 1). To ensure robust operation we'd like to make the noise margins as large as possible.



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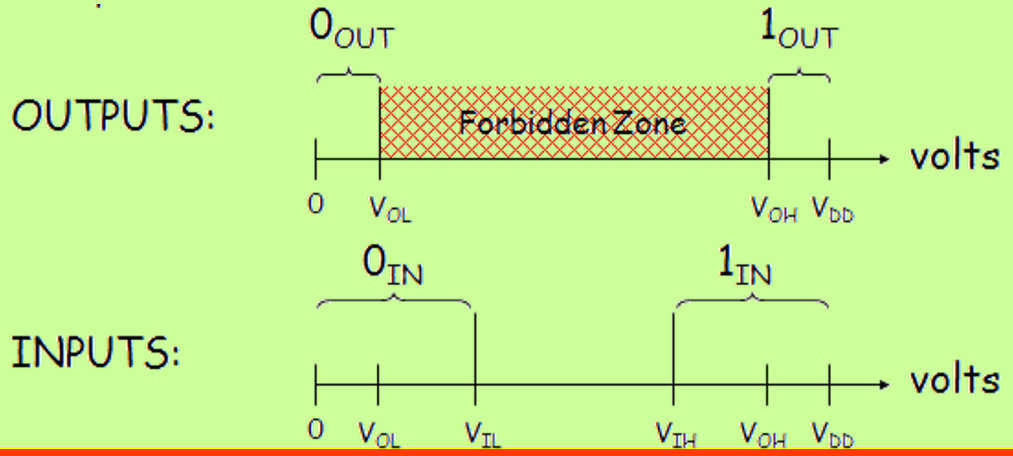
Digital Signaling Specification

Digital input: $V_{IN} < V_{IL}$ or $V_{IN} > V_{IH}$

Digital output: $V_{OUT} < V_{OL}$ or $V_{OUT} > V_{OH}$

Noise margins: $V_{IL} - V_{OL}$ and $V_{OH} - V_{IH}$

Where V_{OL}, V_{IL}, V_{IH} and V_{OH} are part of the specification for a particular family of digital components.



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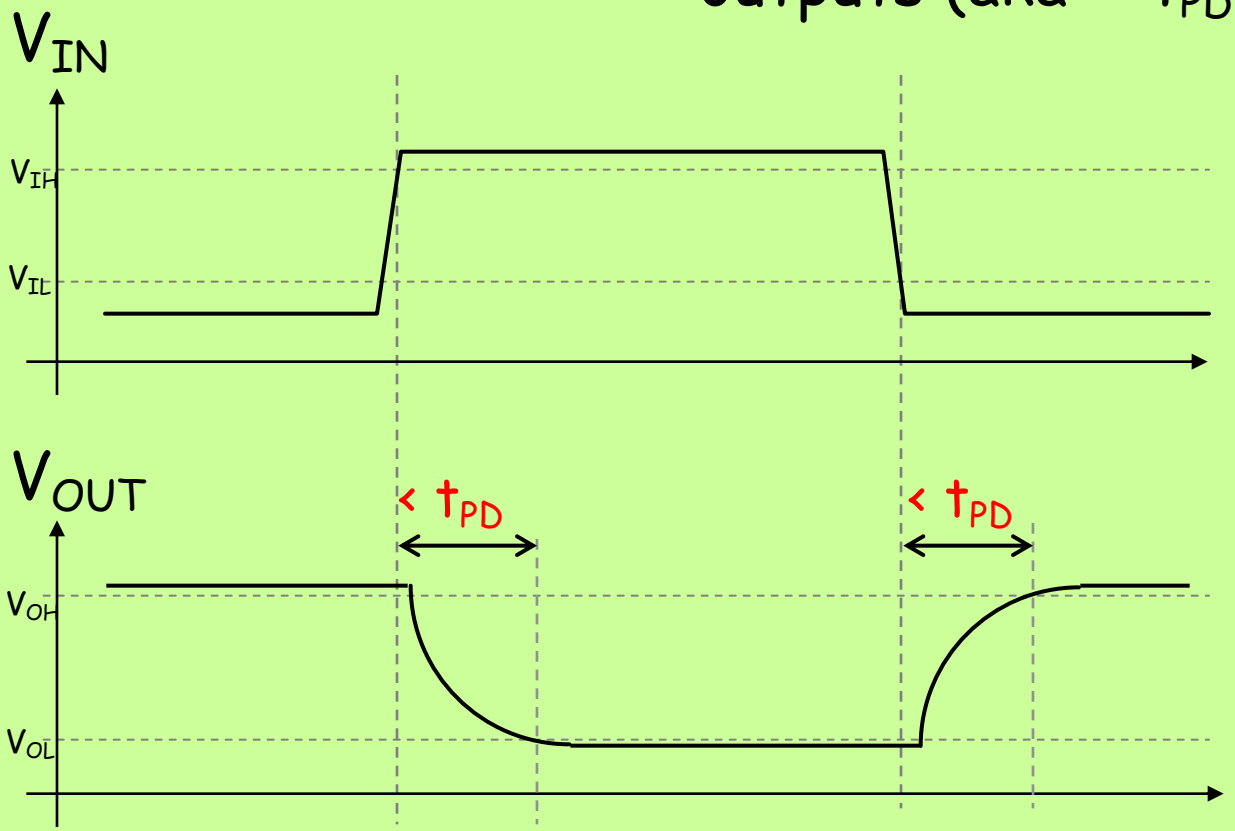
Sample DC (signaling) Specification

I/O Standard	V_{IL}		V_{IH}		V_{OL}	V_{OH}	I_{OL}	I_{OH}
	V, Min	V, Max	V, Min	V, Max	V, Max	V, Min	mA	mA
LVTTL	-0.3	0.8	2.0	3.45	0.4	2.4	Note(3)	Note(3)
LVCMOS33, LVDCI33	-0.3	0.8	2.0	3.45	0.4	V_{cco}-0.4	Note(3)	Note(3)
LVCMOS25, LVDCI25	-0.3	0.7	1.7	V_{cco}+0.3	0.4	V_{cco}-0.4	Note(3)	Note(3)
LVCMOS18, LVDCI18	-0.3	35% V_{cco}	65% V_{cco}	V_{cco}+0.3	0.45	V_{cco}-0.45	Note(4)	Note(4)
LVCMOS15, LVDCI15	-0.3	35% V_{cco}	65% V_{cco}	V_{cco}+0.3	25% V_{cco}	75% V_{cco}	Note(4)	Note(4)
LVCMOS12	-0.3	35% V_{cco}	65% V_{cco}	V_{cco}+0.3	25% V_{cco}	75% V_{cco}	Note(6)	Note(6)
PCI33_3(5)	-0.2	30% V_{cco}	50% V_{cco}	V_{cco}	10% V_{cco}	90% V_{cco}	Note(5)	Note(5)
PCI66_3(5)	-0.2	30% V_{cco}	50% V_{cco}	V_{cco}	10% V_{cco}	90% V_{cco}	Note(5)	Note(5)
PCI-X(5)	-0.2	35% V_{cco}	50% V_{cco}	V_{cco}	10% V_{cco}	90% V_{cco}	Note(5)	Note(5)

Source: Xilinx Virtex 5 Datasheet

Timing Specifications

Propagation delay (t_{PD}): An *upper bound* on the delay from valid inputs to valid outputs (aka "$t_{PD,MAX}$")



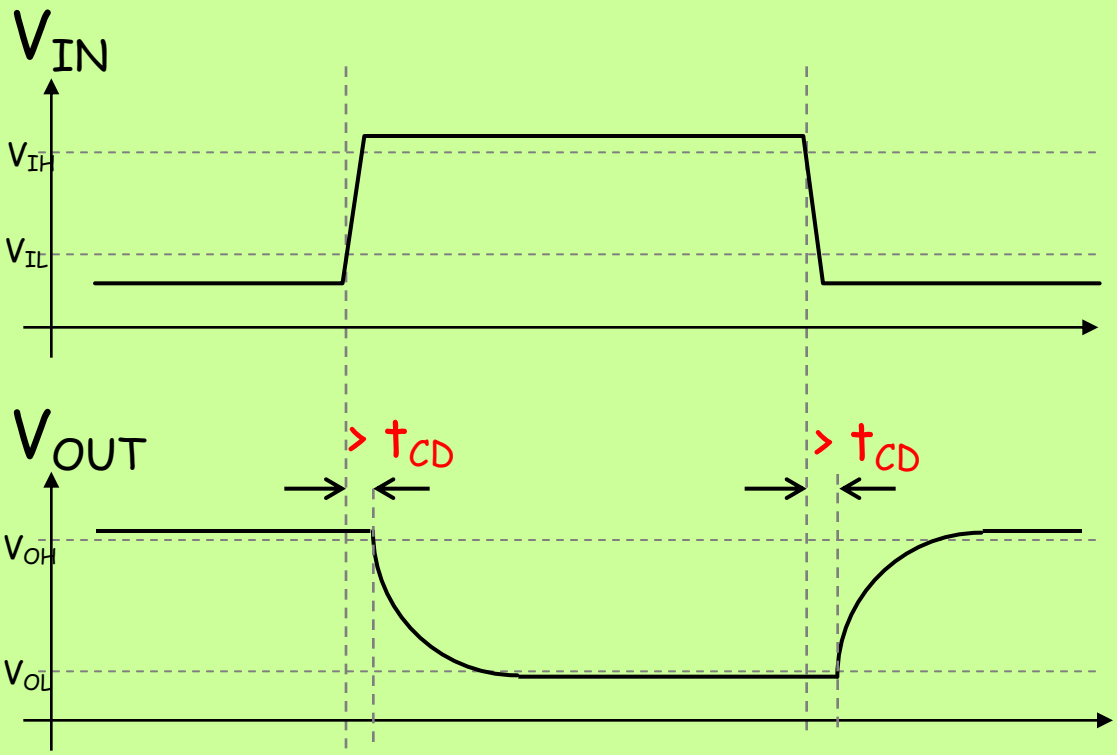
Design goal: *minimize* propagation delay

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Timing Specifications

Contamination delay(t_{CD}): A *lower bound* on the delay from invalid inputs to invalid outputs (aka "$t_{PD,MIN}$")

(an optional, additional timing spec)



Do we really need t_{CD}?

Usually not… it'll be important when we design circuits with registers (coming soon!)

If t_{CD} is not specified, safe to assume it's 0.

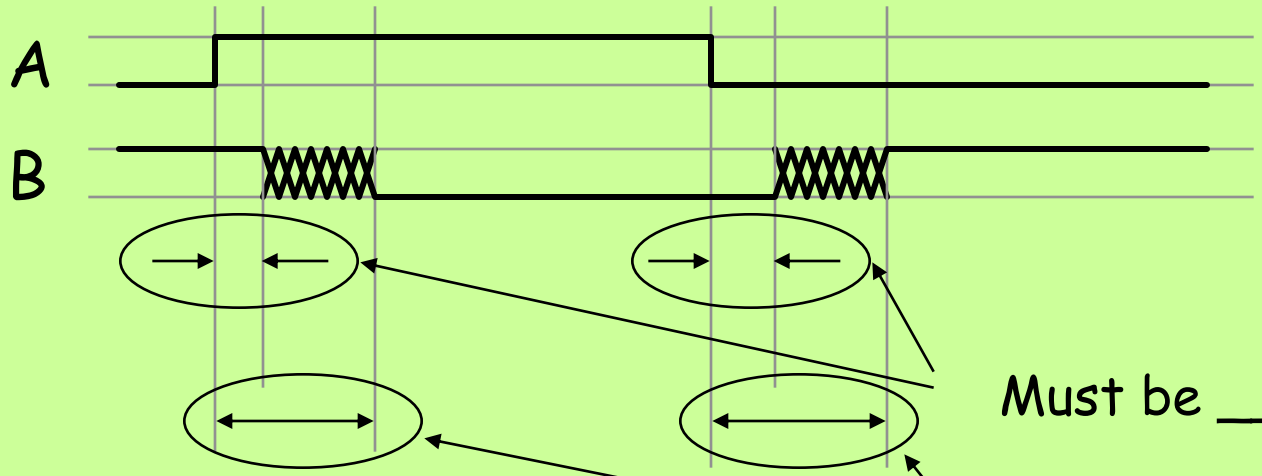
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The Combinational Contract



A	B
0	1
1	0

t_{PD} propagation delay
t_{CD} contamination delay



Note:

1. *No Promises* during ≋
2. Default (conservative) spec: $t_{CD} = 0$

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From transistor to processor

- Transistor
- Digital Logic Gates
- Combinational Logic
  - Car alarm example
- Sequential Logic Example
  - Car alarm example
- Microprocessor

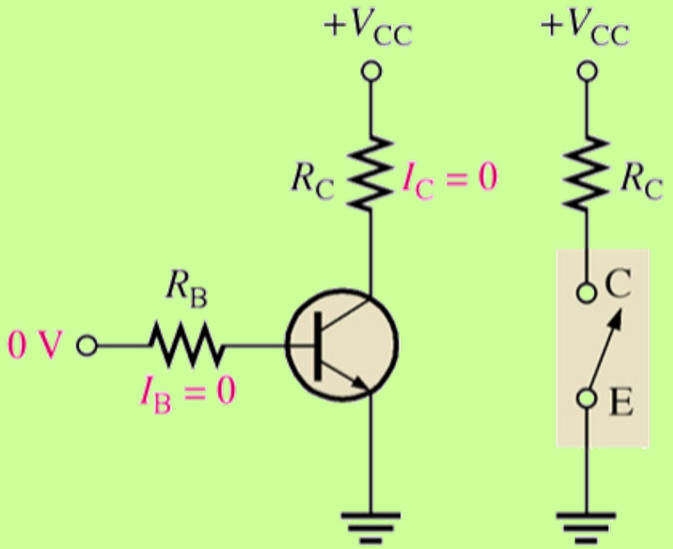
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Transistors

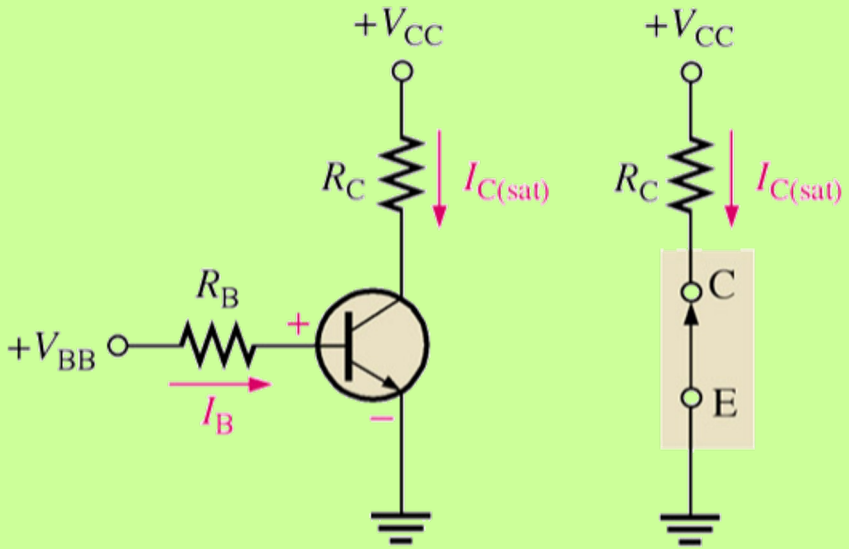
- The fundamental building blocks for all digital logic circuits
- The logic gates are built using transistors
- Electronic binary switches that can be turned on and off
- The 1's and 0's that we talk about are used to represent the on and off states of a transistor

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Transistor = switch



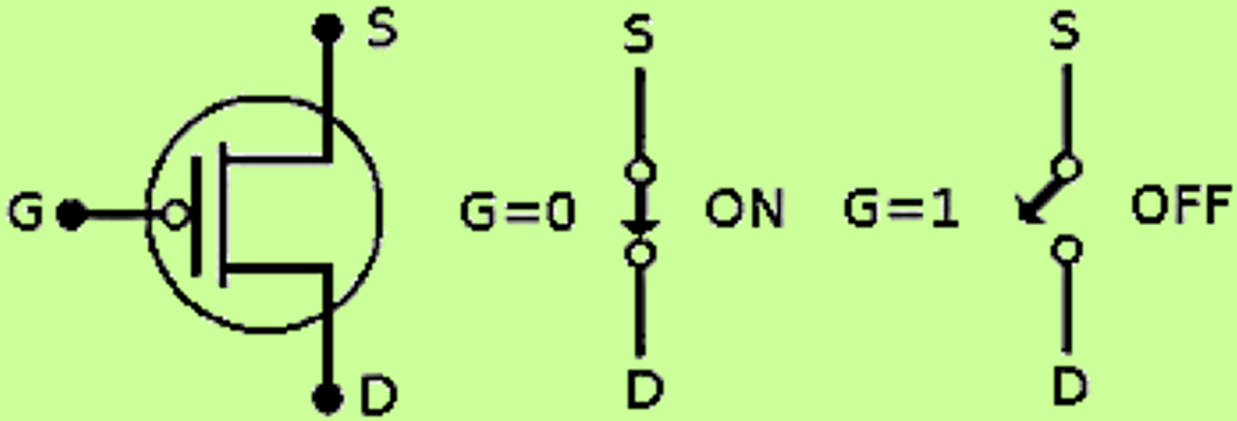
(a) Cutoff — open switch



(b) Saturation — closed switch

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MOSFET=switch

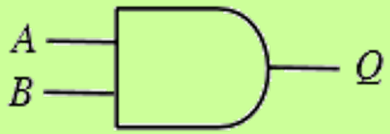


Logic Gates

- All digital logic circuits, whether combinational or sequential, are made up of the three basic logic gates:
    1. AND gate
    2. OR gate
    3. NOT gate
- From these three basic gates, the most powerful computer can be made

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Logic Gates



AND

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

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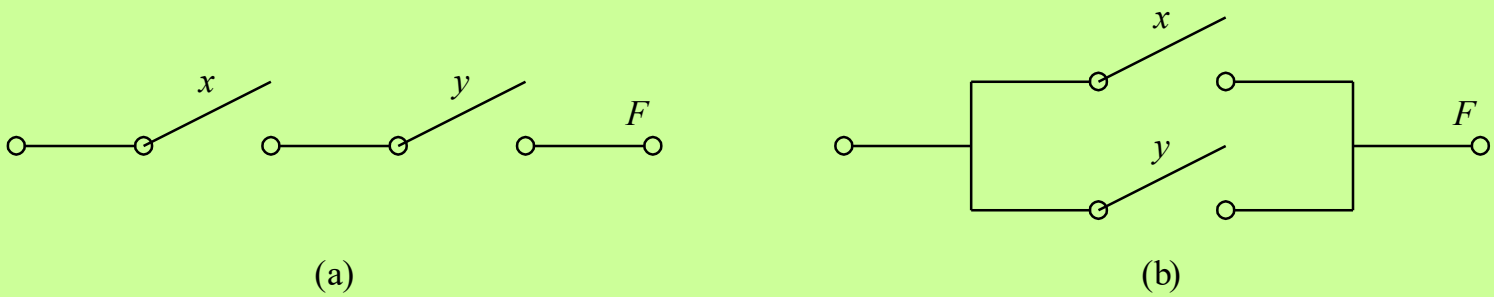
Logic Gates

x	0
y	0
AND: $x \cdot y$	0
OR: $x + y$	0
NOT: x'	1

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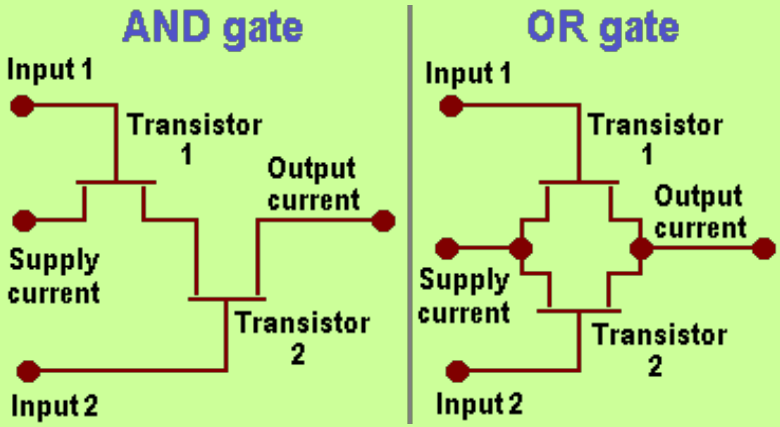
From on-off to logic



- AND F=x and y F=x•y F=xy
- OR F=x or y F=x+y
- NOT F=x ′

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From on-off to logic



- 所有的数字、逻辑运算都可以表示成与、或、非这三个最基本逻辑的组合。
- 这构成了整个数字电路与系统的基础。

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Digital Logic Circuit

- All digital logic circuits are categorized as either:

    1. Combinational circuit
    2. Sequential circuit

- Either Combinational circuit or Sequential circuit are build from logic gate

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Car Security System – Combinational logic

- **Input: M = Siren Manual switch**

  D = Door open

  V = Vibration sensor

  Output: S = Siren

Truth Table

M	D	V	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

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Car Security System – Combinational logic

- **Input: M = Siren Manual switch**

  D = Door open

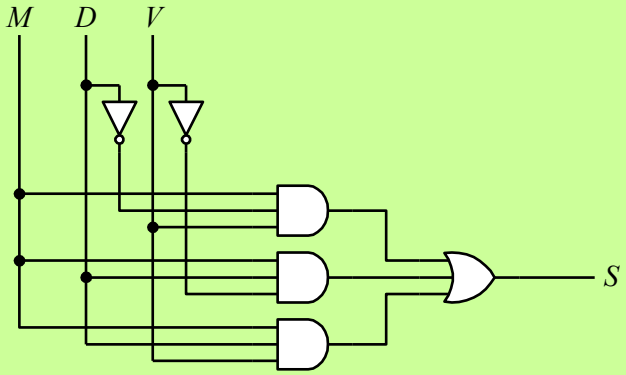
  V = Vibration sensor

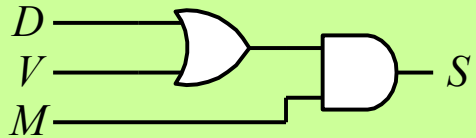
  Output: S = Siren

S= (M D' V) + (M D V') + (M DV)
= M (D' V + D V' + D V)
= M (D' V + D V' + D V + DV)
= M (D(V' + V) + V(D' + D))
= M (D(1) + V(1))
= M (D + V)

M	D	V	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Truth Table





Schematic

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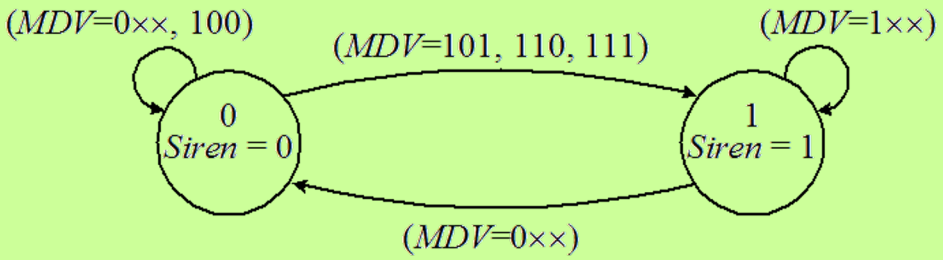
Car Security System – Sequential logic

- **Input: M = Siren Manual switch**

  D = Door open

  V = Vibration sensor

  Output: S = Siren



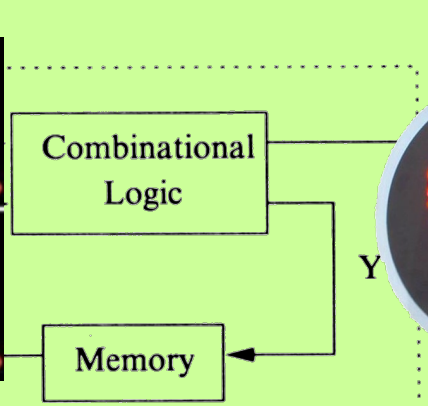
Current State Q_0	Next State Q_{0next}							
	M, D, V							
	000	001	010	011	100	101	110	111
0	0	0	0	0	0	1	1	1
1	0	0	0	0	1	1	1	1

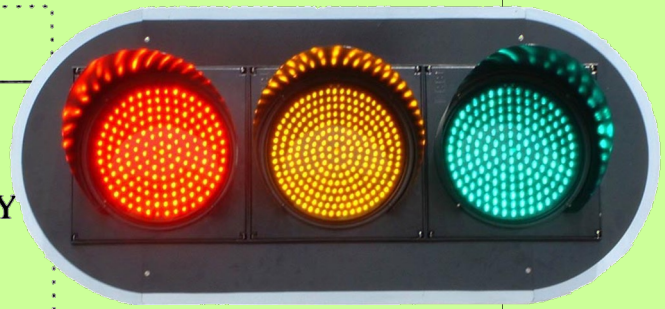
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Combinational vs Sequential

- **Combinational Logic**
  - No state
  - No memory
  - E.g. 7-segment LED
- **Sequential Logic**
  - State
  - Memory
  - E.g. traffic light

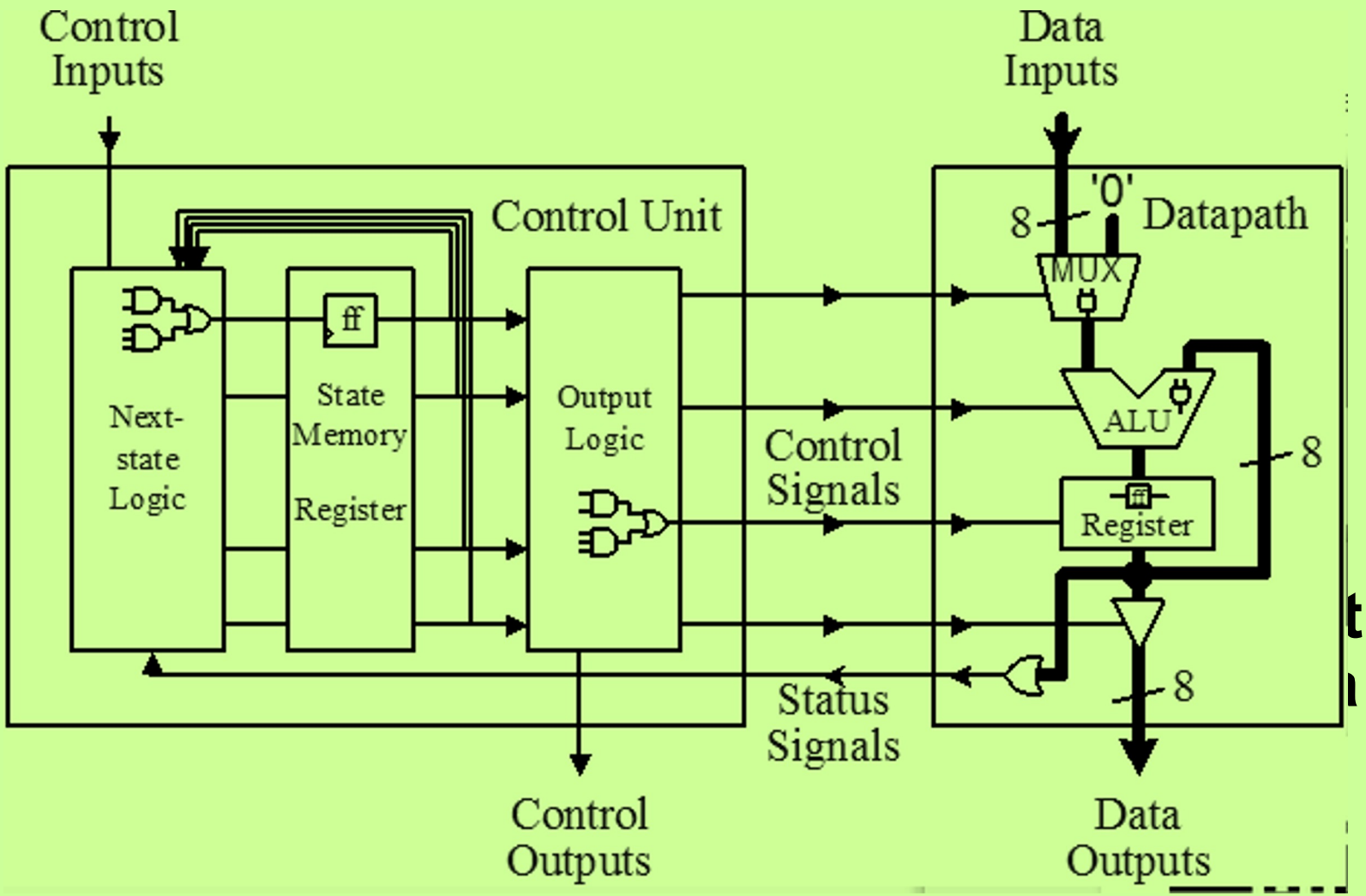






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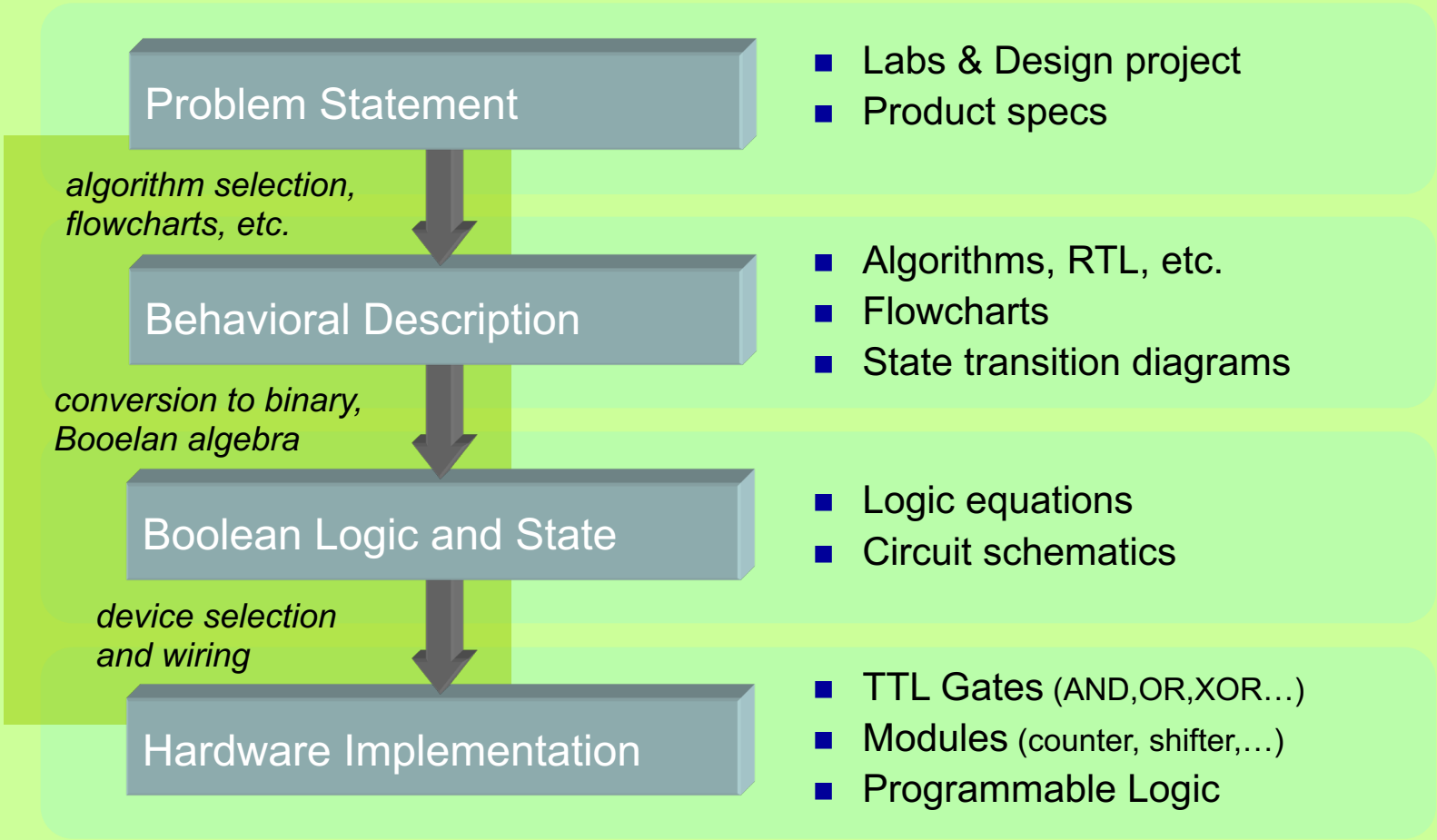
Processor design



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Building Digital Systems

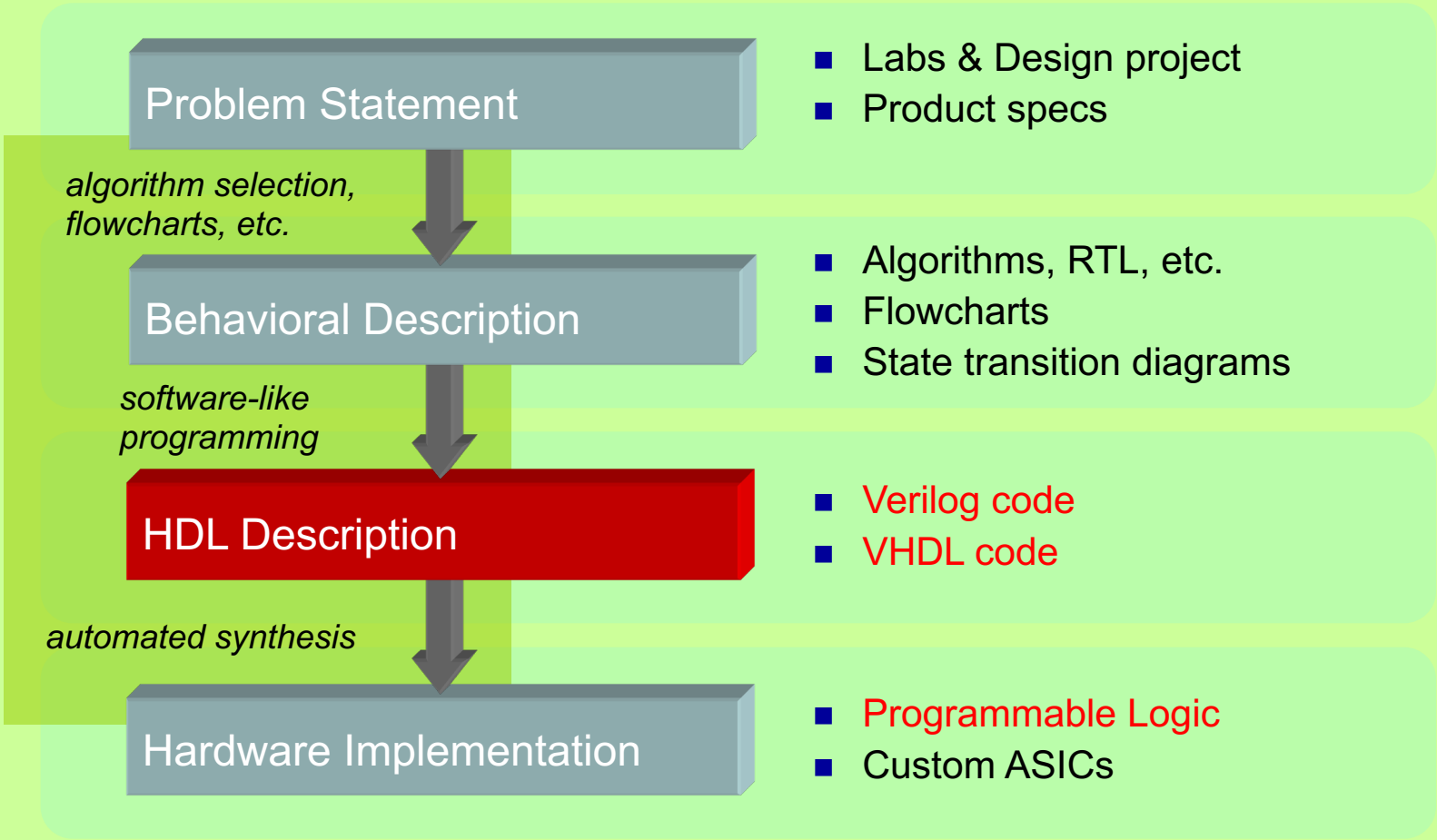
- **Building binary digital solutions to computational problems**



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Building Digital Systems with HDLs

- **Logic synthesis using a Hardware Description Language (HDL) automates the most tedious and error-prone aspects of design**



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开机

检测水箱

检测渣盒

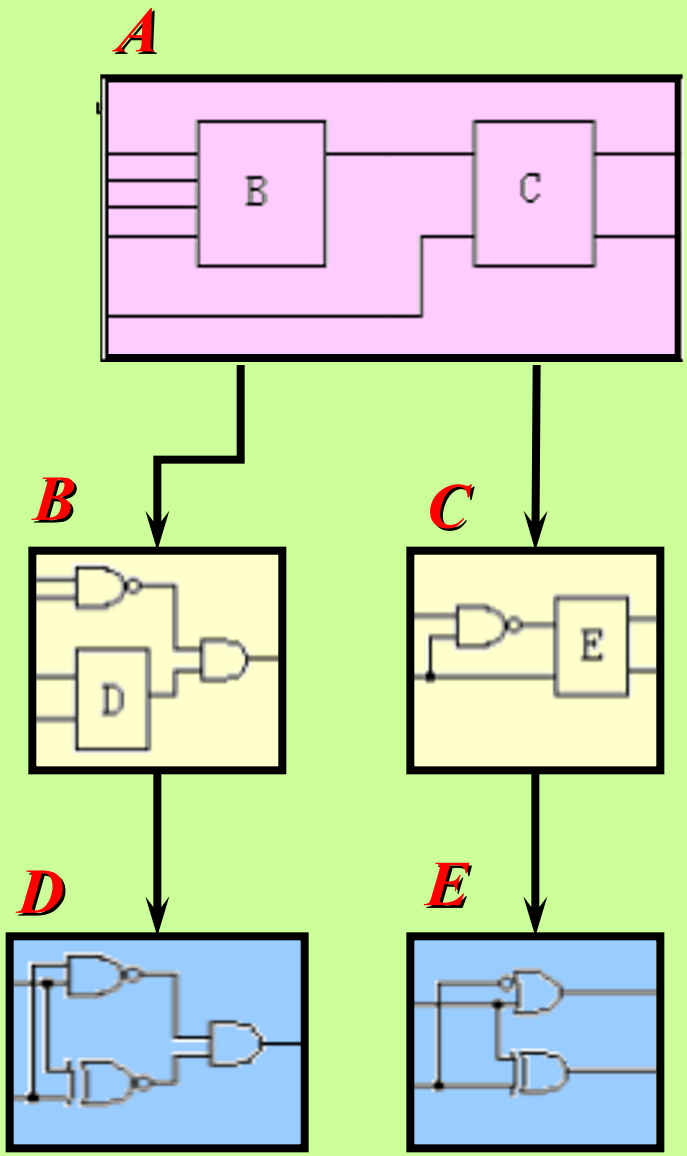
清洗喷嘴

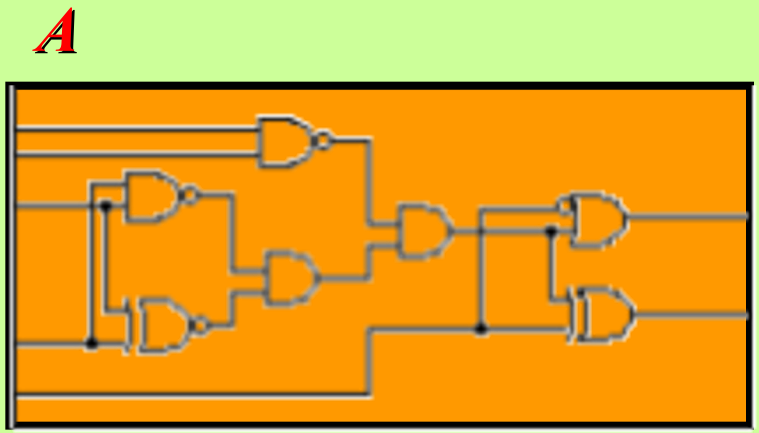
自检结束

等待功能选择



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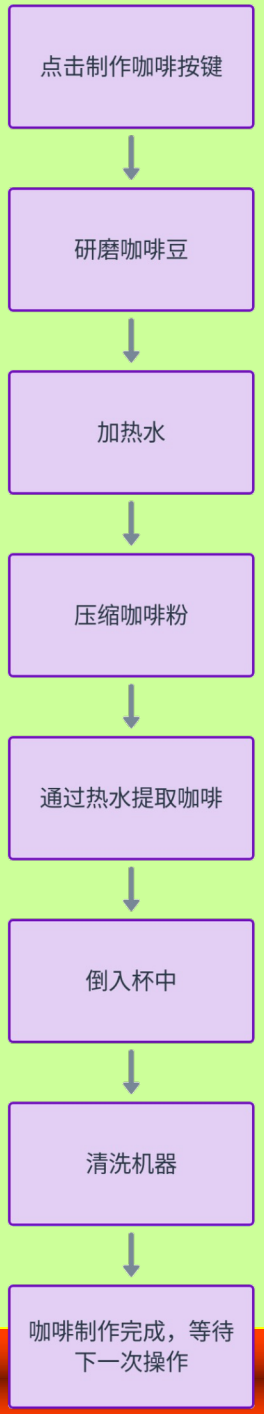




Top-Bottom Design

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- 开关——on/off
- 一杯浓缩
- 两杯浓缩
- 一杯美式
- 两杯美式
- 蒸汽







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Next…..

- Next coming
  - Boolean algebra
  - Combinational Circuit
  - Gate-level Minimization

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